TASK 1

INSY7315

Client Name: Koos Matli Primary School

Women In Tech

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INTRODUCTION

Koos Matli Primary School is a community-focused primary school in Mamelodi committed to providing quality education and creating an environment in which learners can develop academically, socially, and personally. In support of the school's vision, “Thuto ke Lesedi” (Education is Light), this project proposes the development of an integrated digital school platform comprising a public-facing website and a mobile application. The proposed system is intended to address the need for more accessible, efficient, and centralised communication and information management between the school, parents, teachers, learners, and the wider school community.

The proposed website will function as the school's public digital presence, providing prospective parents, existing families, and community members with reliable access to information about the school, including its mission and vision, principal's message, staff, admissions, news and announcements, learner achievements, gallery, events, contact information, and location. The mobile application will serve as the primary digital administration and communication platform, providing role-based functionality for parents, teachers, and school administrators. Parents will be able to access relevant information such as their children's timetables, school announcements, fees, application status, academic information, and school events, while teachers and administrators will have appropriate functionality for managing classes, timetables, announcements, applications, users, and other administrative processes.

The justification for the proposed system is based on the opportunity to improve the accessibility, efficiency, and reliability of the school's communication and administrative processes while considering the specific needs of its community. The application will follow an offline-first and data-conscious approach, allowing appropriate information to remain accessible when connectivity is limited and reducing unnecessary mobile data usage. The solution will also provide a foundation for future multilingual support, allowing the platform to become more accessible to South African users. As the system will process personal and potentially sensitive information relating to learners, parents, teachers, and school administration, security, role-based access control, privacy, and compliance with the Protection of Personal Information Act (POPIA) will form an integral part of the system design.

This document presents the proposed system from a complete software development and management perspective. The subsequent sections define the functional and non-functional requirements, identify the user roles and user stories, and document the intended user experience through journey mapping. The project's work breakdown structure establishes how the solution will be planned and delivered, while the analysis and design artefacts provide models of the system's domain, classes, interactions, states, data, and architecture. The document further evaluates the choice of technology, database and data schema, design and architecture patterns, cloud architecture, security controls, DevOps practices, and deployment approach. Finally, running costs and change management are considered to assess the system's long-term feasibility, sustainability, and ability to adapt to the school's future requirements. Collectively, these considerations provide management with a structured justification for

the proposed solution and demonstrate how the system can deliver practical value to Koos Matli Primary School while remaining secure, maintainable, scalable, and aligned with the needs of its users.

REQUIREMENTS

USER ROLES

The proposed system has been designed to accommodate different categories of users, each with specific responsibilities, access requirements, and interactions with the system. Role-based access will ensure that users are only able to access functionality and information relevant to their responsibilities, supporting both usability and the protection of personal information (Satzinger et al, 2016). These are the user roles:

- **Public Visitor:** A prospective parent, existing parent, learner, community member, or other member of the public who accesses the school's website without logging into the system. Public Visitors can access general school information. Public Visitors will not have access to private learner, parent, teacher, financial, or administrative information.
- **Parent/Guardian:** An authenticated user of the mobile application who is responsible for one or more learners attending or applying to Koos Matli Primary School. Parents/Guardians can access information relevant to their children and other approved school communications. Access to learner information will be restricted so that a Parent/Guardian can only view information relating to learners for whom they are authorised as a parent or guardian.
- **Teacher:** An authenticated school staff member responsible for teaching learners and managing academic or class-related information within the system. Teachers can access their assigned classes and timetables and can create or distribute appropriate announcements, homework, learning materials, and other information to authorised users. Teachers will only have access to learner and class information necessary for their assigned responsibilities.
- **Administrator:** An authorised member of the school's administrative or management team responsible for managing the overall system and school information. Administrators have elevated privileges that allow them to manage users, classes, applications, timetables, announcements, staff information, school information, and other administrative functions. Administrative access will be restricted and protected through appropriate authentication, authorisation, and security controls.

USER STORIES

The following user stories have been developed to describe the key interactions between each identified user role and the proposed system (Satzinger et al, 2016). The stories follow the standard format:

As a [user role], I want to [action], so that [reason/benefit].

Public Visitor User Stories

1. As a Public Visitor, I want to view information about the school, so that I can understand the school's purpose, values, and educational environment.
2. As a Public Visitor, I want to view staff information, so that I can learn about the school's teaching and administrative team.
3. As a Public Visitor, I want to view admissions information, so that I can understand how to apply for a learner to attend the school.

Parent/Guardian User Stories

1. As a Parent/Guardian, I want to securely log into the mobile application, so that I can access information relating to myself and my child or children.
2. As a Parent/Guardian, I want to view my child's timetable, so that I can know which classes and activities they have during the school day.
3. As a Parent/Guardian, I want to track my child's admission application, so that I know the status of the application without having to contact the school for every update.

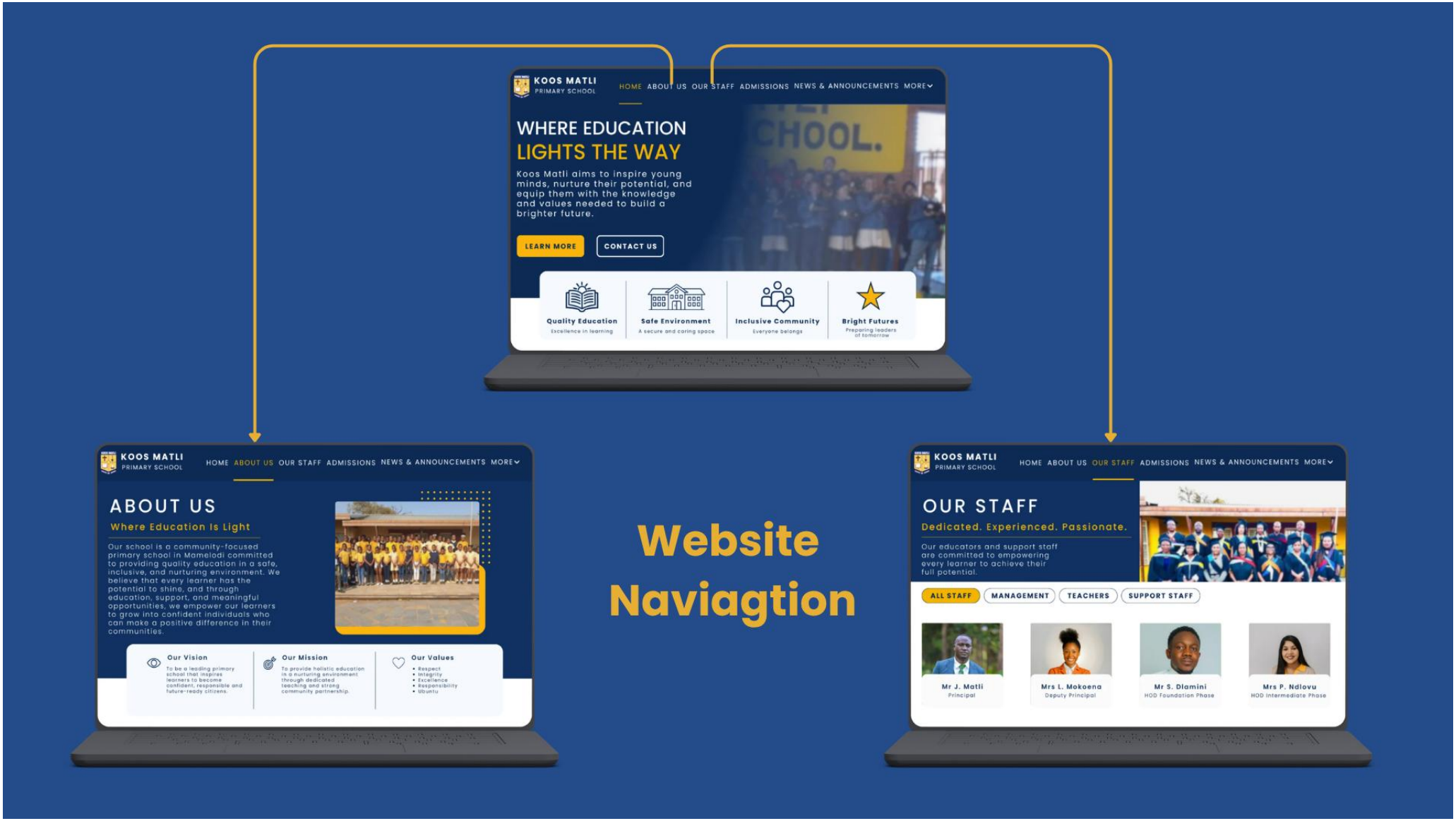
Teacher User Stories

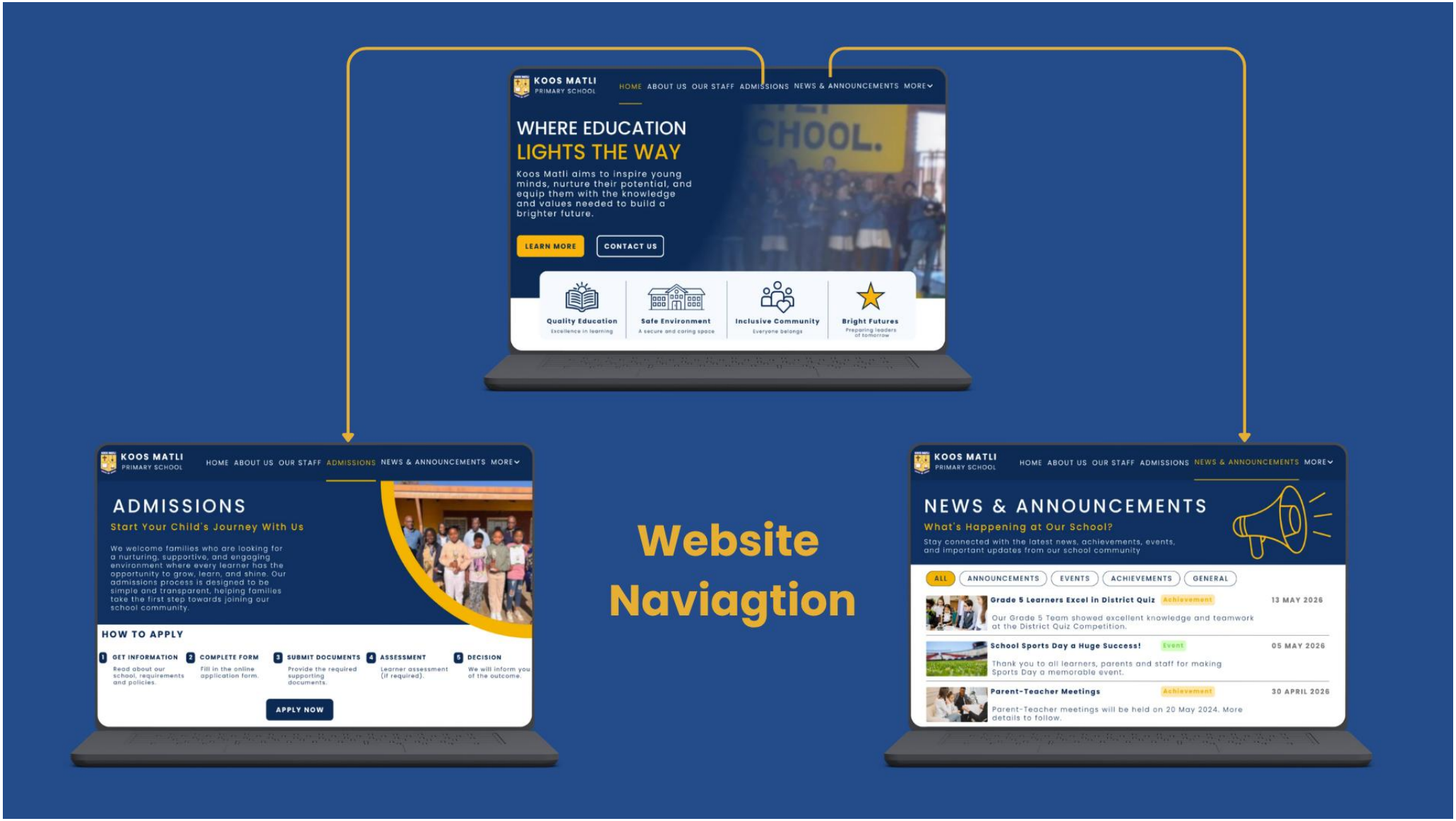
1. As a Teacher, I want to securely log into the application, so that I can access the teaching and class-management functions assigned to me.
2. As a Teacher, I want to view my teaching timetable, so that I know which classes and activities I am responsible for each day.
3. As a Teacher, I want to upload homework or learning materials, so that learners and parents can access relevant educational resources.

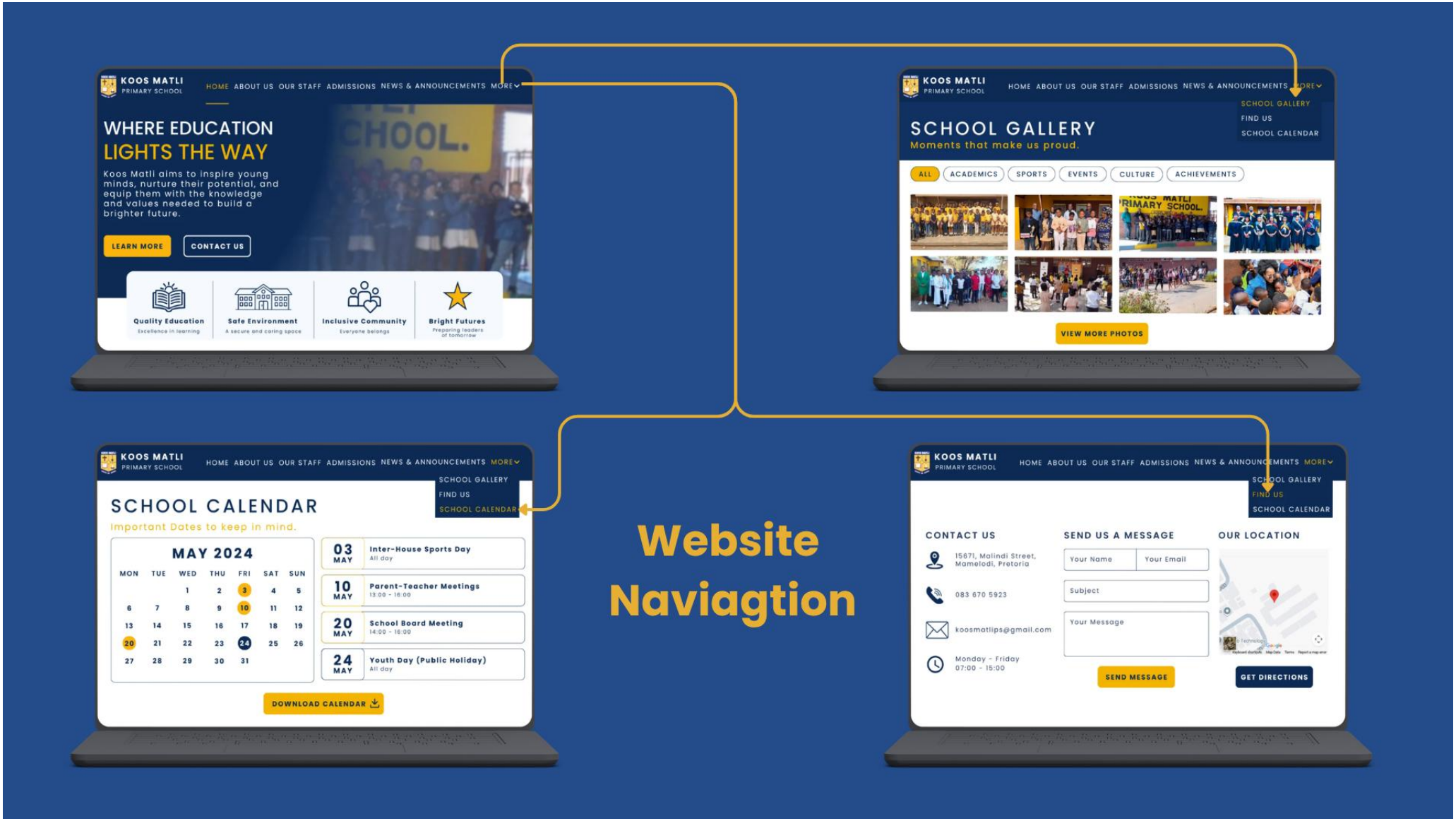
Administrator User Stories

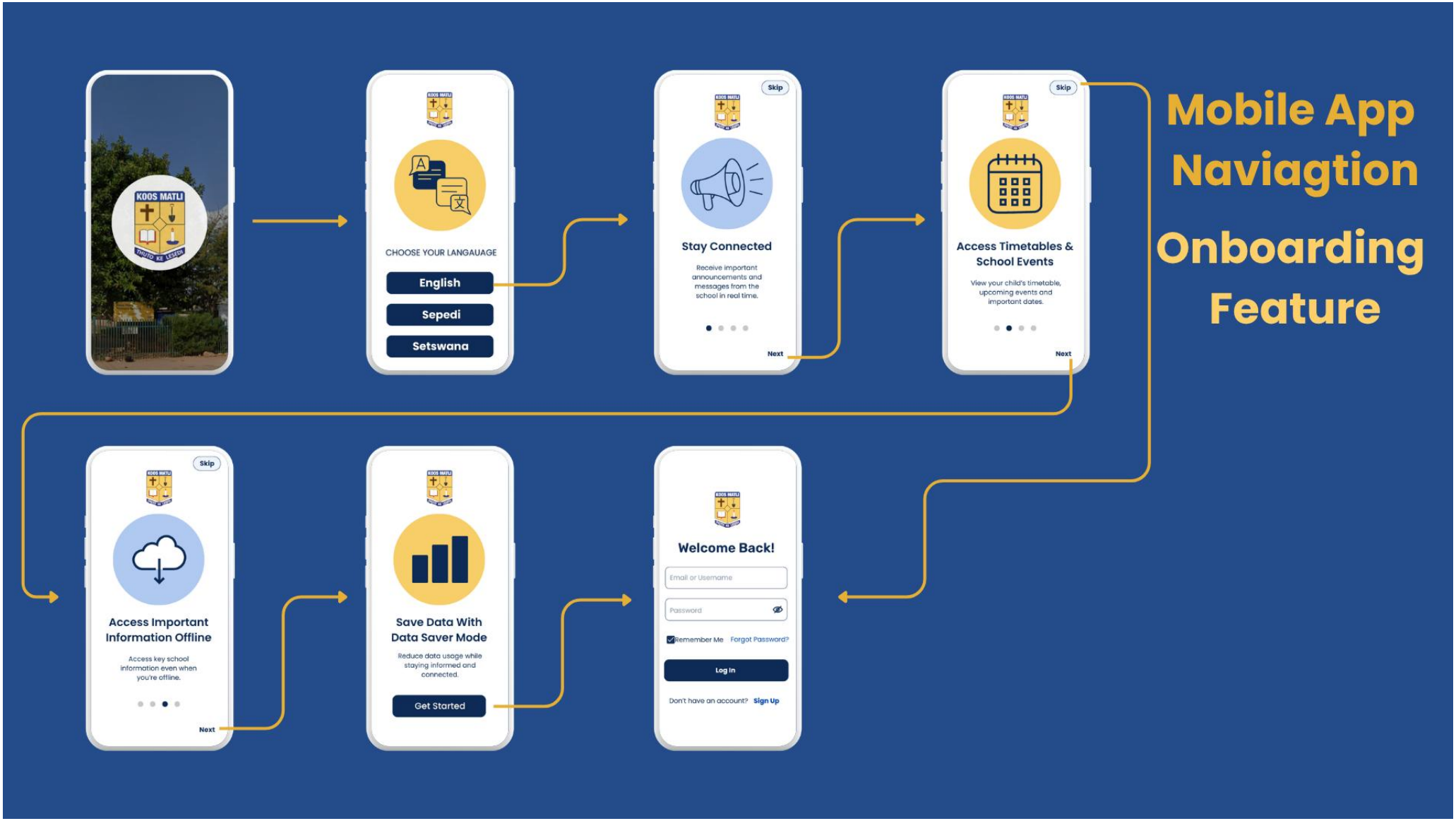
1. As an Administrator, I want to securely log into the administration system, so that I can manage school information and system functions.
2. As an Administrator, I want to manage parent, teacher, and learner accounts, so that authorised users can access the appropriate system functionality.
3. As an Administrator, I want to assign appropriate user roles and permissions, so that users can only access information relevant to their responsibilities.

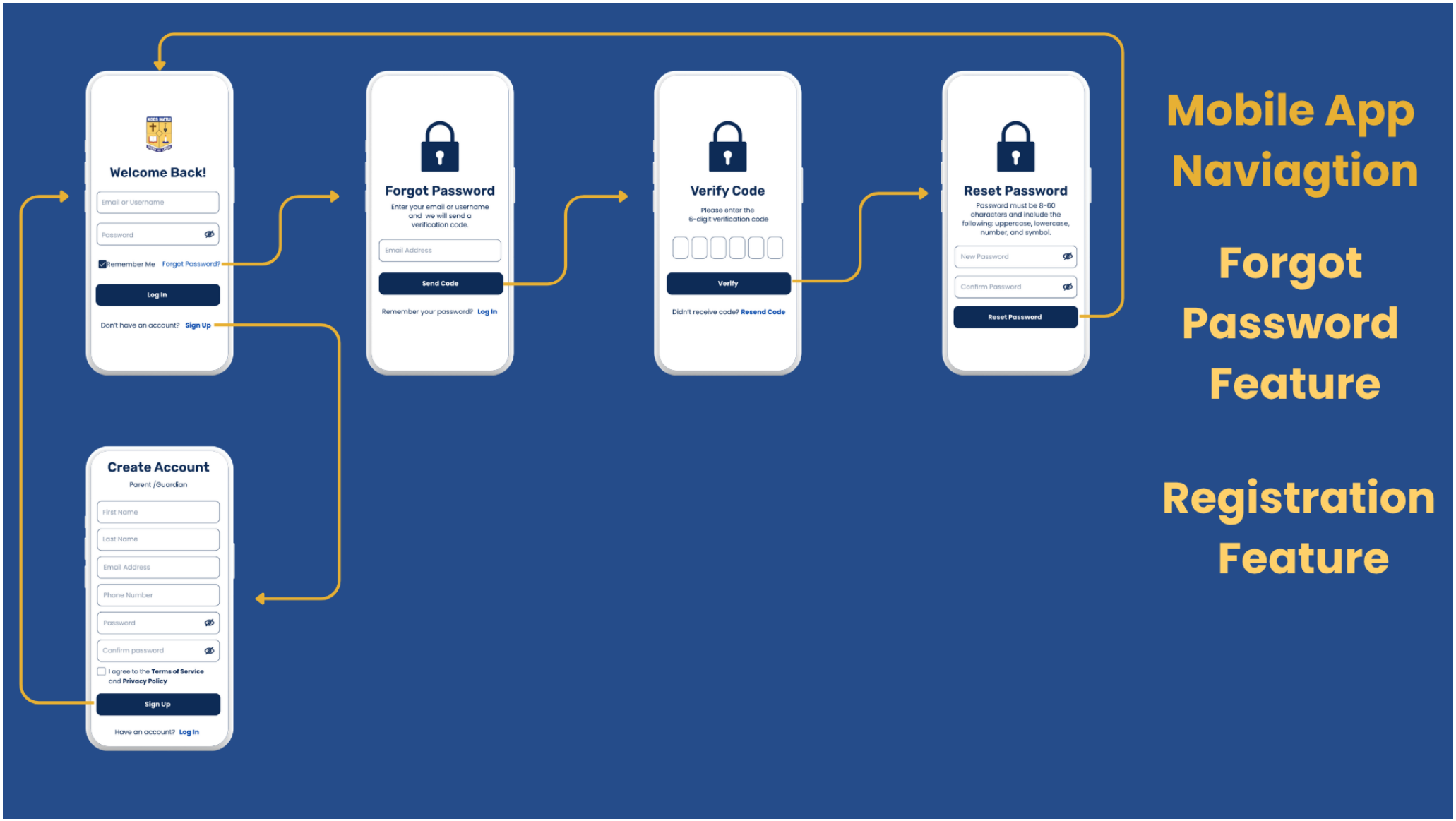
UX JOURNEY MAP

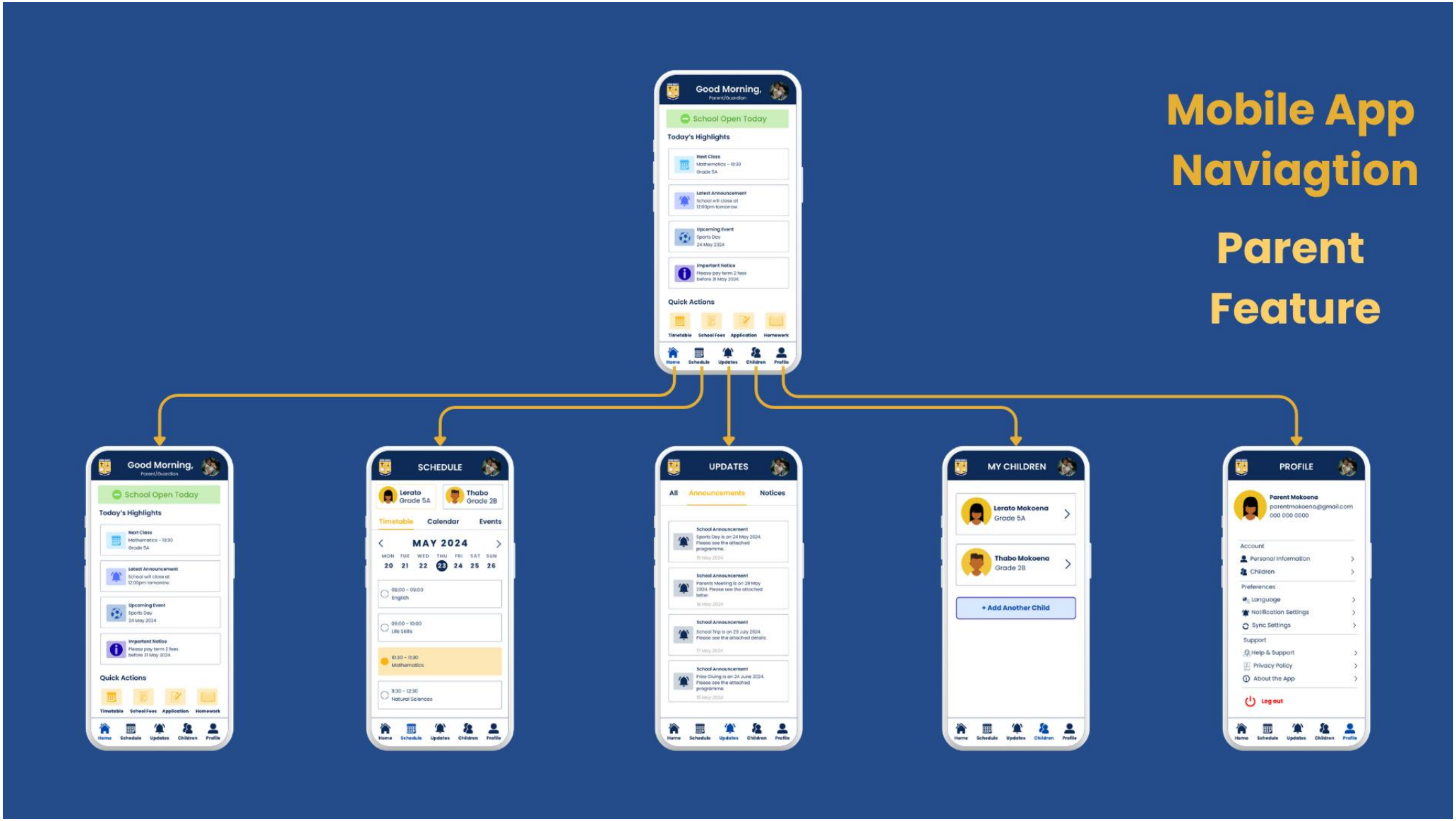


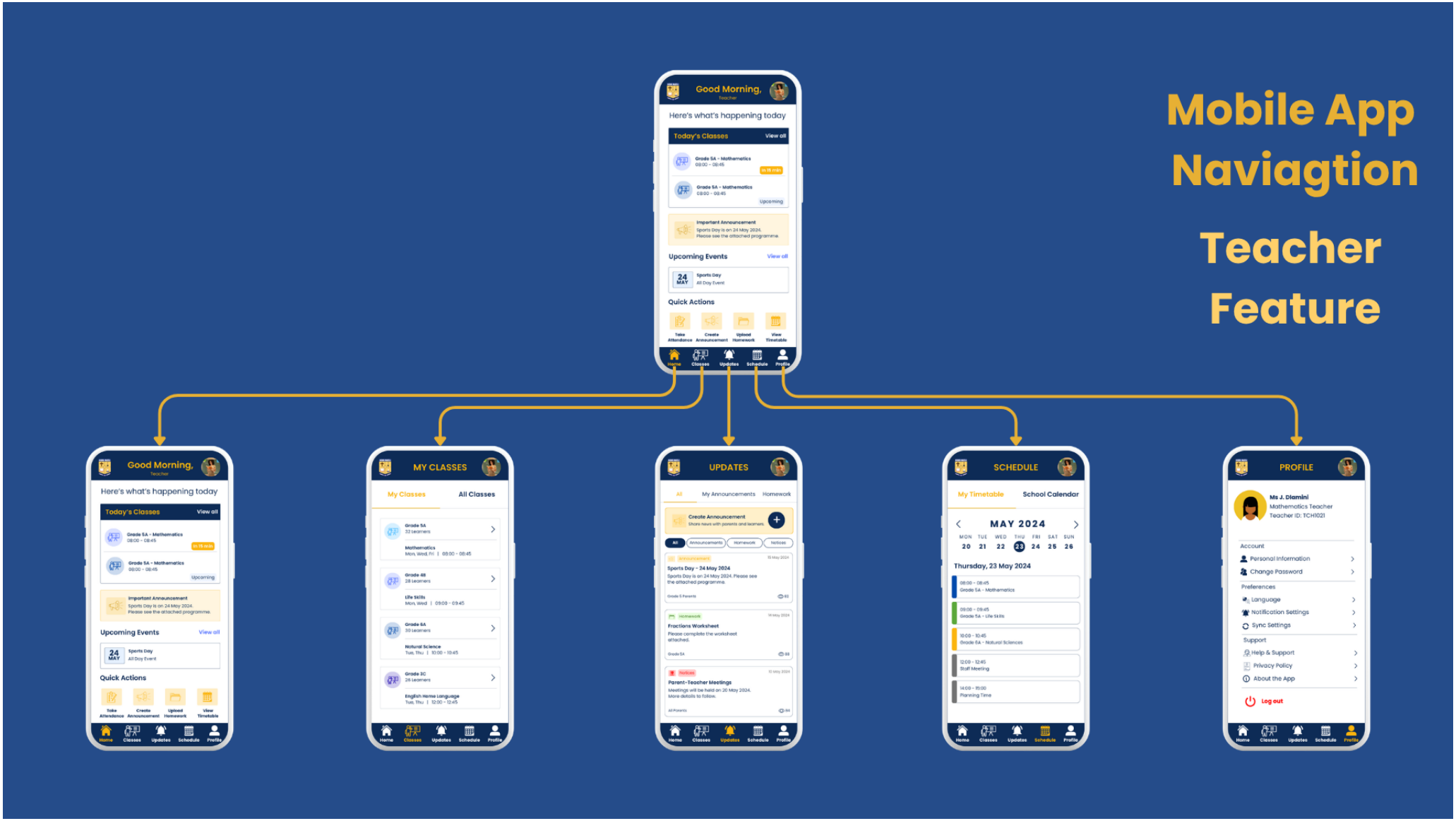

KOOS MATLI
PRIMARY SCHOOL
HOME ABOUT US OUR STAFF ADMISSIONS NEWS & ANNOUNCEMENTS MORE
WHERE EDUCATION
LIGHTS THE WAY
Koos Matli aims to inspire young minds, nurture their potential, and equip them with the knowledge and values needed to build a brighter future.
LEARN MORE
CONTACT US
Quality Education
Excellence in learning
Safe Environment
A secure and caring space
Inclusive Community
Everyone belongs
Bright Futures
Preparing leaders of tomorrow
ADMISSIONS
Start Your Child's Journey With Us
We welcome families who are looking for a nurturing, supportive, and engaging environment where every learner has the opportunity to grow, learn, and shine. Our admissions process is designed to be simple and transparent, helping families take the first step towards joining our school community.
HOW TO APPLY
1 GET INFORMATION
Read about our school, requirements and policies.
2 COMPLETE FORM
Fill in the online application form.
3 SUBMIT DOCUMENTS
Provide the required supporting documents.
4 ASSESSMENT
Learner assessment (if required).
5 DECISION
We will inform you of the outcome.
APPLY NOW
Website
Naviagtion
NEWS & ANNOUNCEMENTS
What's Happening at Our School?
Stay connected with the latest news, achievements, events, and important updates from our school community
ALL
ANNOUNCEMENTS
EVENTS
ACHIEVEMENTS
GENERAL
Grade 5 Learners Excel in District Quiz
Achievement
13 MAY 2026
Our Grade 5 Team showed excellent knowledge and teamwork at the District Quiz Competition.
School Sports Day a Huge Success!
Event
05 MAY 2026
Thank you to all learners, parents and staff for making Sports Day a memorable event.
Parent-Teacher Meetings
Achievement
30 APRIL 2026
Parent-Teacher meetings will be held on 20 May 2024. More details to follow.


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Inclusive Community
Everyone belongs
Bright Futures
Preparing leaders of tomorrow
SCHOOL GALLERY
FIND US
SCHOOL CALENDAR
SCHOOL GALLERY
Moments that make us proud.
ALL
ACADEMICS
SPORTS
EVENTS
CULTURE
ACHIEVEMENTS
VIEW MORE PHOTOS
SCHOOL CALENDAR
Important Dates to keep in mind.
MAY 2024
MON TUE WED THU FRI SAT SUN
03 MAY Inter-House Sports Day All day
10 MAY Parent-Teacher Meetings 13:00 - 16:00
20 MAY School Board Meeting 14:00 - 16:00
24 MAY Youth Day (Public Holiday) All day
DOWNLOAD CALENDAR
Website
Naviagtion
CONTACT US
15671, Malindi Street, Mamelodi, Pretoria
083 670 5923
koosmatlips@gmail.com
Monday - Friday 07:00 - 15:00
SEND US A MESSAGE
Your Name
Your Email
Subject
Your Message
SEND MESSAGE
OUR LOCATION
GET DIRECTIONS


Mobile App Naviagtion
Onboarding Feature
KOOS MATLI
MOOTO KE LESEDI
CHOOSE YOUR LANGAUAGE
English
Sepedi
Setswana
Skip
Stay Connected
Receive important announcements and messages from the school in real time.
Next
Skip
Access Timetables & School Events
View your child's timetable, upcoming events and important dates.
Next
Skip
Access Important Information Offline
Access key school information even when you're offline.
Next
Save Data With Data Saver Mode
Reduce data usage while staying informed and connected.
Get Started
Welcome Back!
Email or Username
Password
Remember Me
Forgot Password?
Log In
Don't have an account? Sign Up


Welcome Back!
Email or Username
Password
Remember Me
Forgot Password?
Log In
Don't have an account? Sign Up
Forgot Password
Enter your email or username and we will send a verification code.
Email Address
Send Code
Remember your password? Log In
Verify Code
Please enter the 6-digit verification code
Verify
Didn't receive code? Resend Code
Reset Password
Password must be 8-60 characters and include the following: uppercase, lowercase, number, and symbol.
New Password
Confirm Password
Reset Password
Create Account
Parent /Guardian
First Name
Last Name
Email Address
Phone Number
Password
Confirm password
I agree to the Terms of Service and Privacy Policy
Sign Up
Have an account? Log In
Mobile App Naviagtion
Forgot Password Feature
Registration Feature


Mobile App
Naviagtion
Parent
Feature
Good Morning,
School Open Today
Today's Highlights
Quick Actions
SCHEDULE
UPDATES
MY CHILDREN
PROFILE
Lerato Mokoena
Grade 5A
Thabo Mokoena
Grade 2B
+ Add Another Child
Log out


Mobile App
Naviagtion
Teacher
Feature
Good Morning,
Here's what's happening today
Today's Classes
View all
Upcoming Events
Quick Actions
MY CLASSES
UPDATES
SCHEDULE
PROFILE


Mobile App
Naviagtion
Admin
Feature
Good Morning,
School Overview
356
42
128
Today's Summary
New Applications 3
Pending Approvals 5
Announcements Sent 2
View Reports
USERS
Parents
Teachers
Learners
Add New User
APPLICATIONS
Add New User
STAFF & CLASS
Add Staff Member
PROFILE
Account Settings
Roles & Permissions
Security
Backup & Restore
Audit Logs
System Settings

Work Breakdown Structure (Gido et al, 2022)

Level	WBS	Task Name	Task Description	Task Duration	Predecessor(s)	Team Member	Resources Needed
1	**Project Planning & Requirements Analysis**						
	1.1	Client Consultation	Meet with the client to understand the school's needs, challenges and expectations for the website and mobile application.	2 days	None	Tlhomamo	Client, meeting platform, meeting notes
	1.2	Requirements Gathering	Identify and document the functional and non-functional requirements for the website and application.	2 days	1.1	Tlhomamo	Client requirements, questionnaires, project brief
	1.3	Project Planning	Define the project scope, deliverables, timelines, responsibilities and development approach.	1 day	1.2	Tlhomamo	WBS, project plan, task management tools
2	**UI/UX Design & Prototyping**						
	2.1	User Experience Design	Create user flows, navigation structures and UX journey maps for the website and different application roles.	3 days	1.3	Moagi	Canva, UX journey maps, requirements
	2.2.	Wireframing	Develop low-fidelity layouts showing the structure and placement of content and interface elements.	3 days	2.1	Moagi	Canva, wireframing tools
	2.3	High-Fidelity Prototypes	Develop detailed website and mobile application mock-ups using the school's branding, colours and approved layouts.	4 days	2.2	Moagi	Canva, school logo, colour palette, images
3	**Front-End Development**						

	3.1	Website Development	Build the responsive school website based on the approved UI/UX designs.	5 days	2.3	Oratile	HTML, CSS, JavaScript, Visual Studio, UI designs
	3.2	Mobile Application Interface Development	Develop the user interfaces for the parent, teacher and administrator sections of the mobile application.	6 days	2.3	Oratile	Android Studio, Kotlin, UI prototypes
	3.3.	Navigation & Responsive Implementation	Implement navigation, forms, buttons, menus and responsive layouts across supported devices.	4 days	3.1, 3.2	Oratile	Development devices, IDE, testing tools
4	**Back-End & Database Development**						
	4.1	Database Design & Implementation	Design and implement the database structure for users, learners, staff, applications, announcements and other system data.	4 days	1.3	Nkumbu	Database management system, ERD, development environment
	4.2	Authentication & Role-Based Access	Implement secure authentication and ensure parents, teachers and administrators can only access authorised functionality and information.	4 days	4.1	Nkumbu	Authentication framework, security tools, database
	4.3.	API & Server-Side Functionality	Develop APIs and backend services that allow the website and mobile application to communicate with the database and process information.	5 days	4.2	Nkumbu	Backend framework, APIs, server/cloud environment
5	**System Integration, Testing & Finalisation**						
	5.1	System Integration	Connect the front-end interfaces with the backend services and database and ensure that the different	3 days	3.3, 4.3	All members	Git/GitHub, development environments, database, API

			components communicate correctly.				
	5.2	Testing & Bug Fixing	Conduct functional, usability, security, navigation and responsive testing and resolve identified issues.	4 days	5.1	All members	Test cases, devices, browsers, debugging tools
	5.3	Final Review & Deployment	Review the completed system against the client's requirements, make final improvements, prepare documentation and deploy/present the final solution.	3 days	5.2	All members	Client feedback, documentation, deployment environment, presentation materials

NON-FUNCTIONAL REQUIREMENTS

The following non-functional requirements were identified through consultation and interaction with the client, Koos Matli Primary School. These requirements define the expected quality, usability, performance, security, accessibility, and operational characteristics of the proposed website and mobile application. They complement the functional requirements by specifying how the system should operate and the standard of service it should provide to the school's users (Satzinger et al, 2016). The requirements have been grouped according to relevant quality attributes to ensure that the proposed solution is reliable, accessible, maintainable, and appropriate for the school's environment:

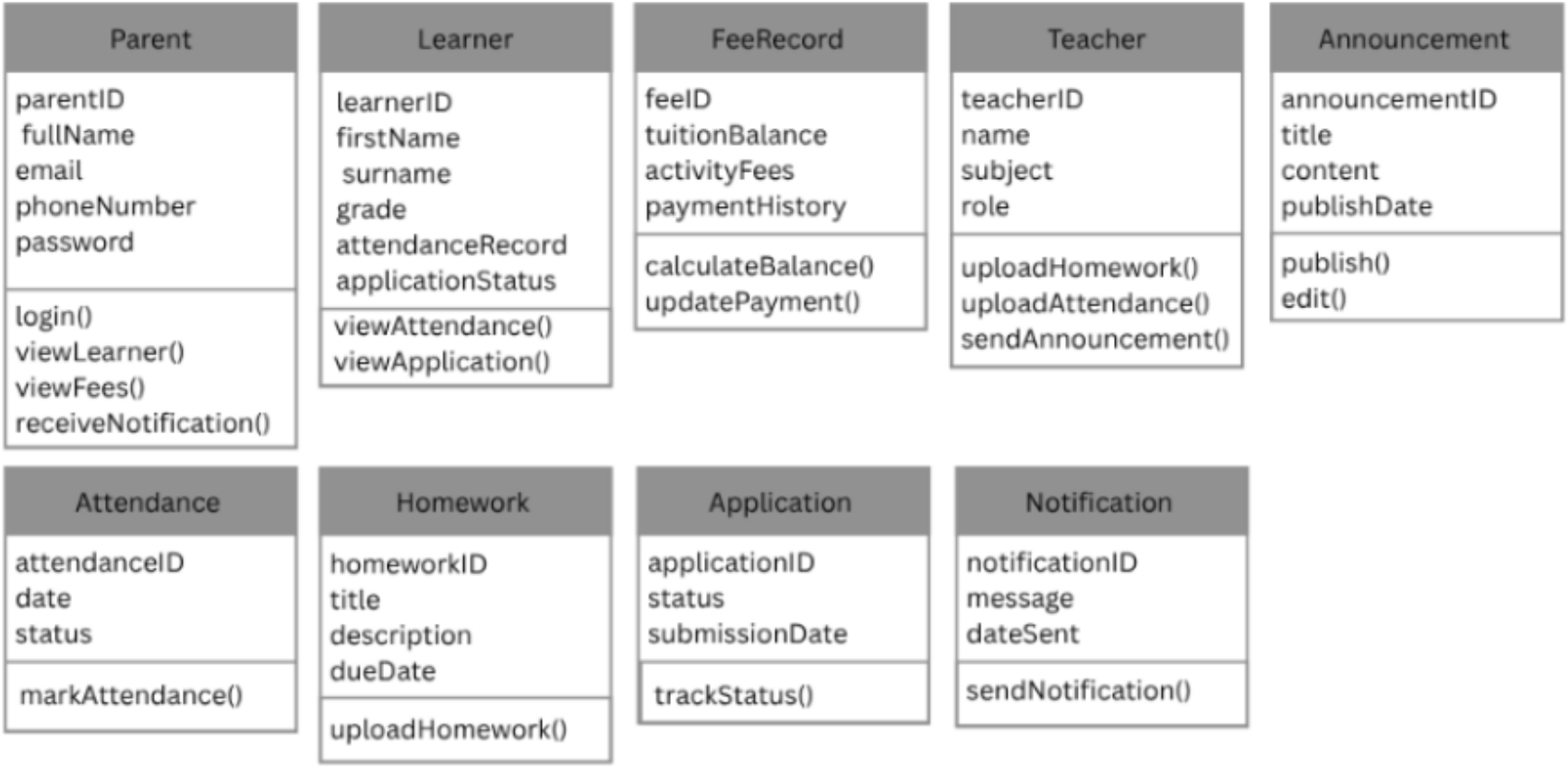
- **Usability** (Satzinger et al, 2016)
  - **Ease of Navigation:** The website and mobile application shall provide a clear and intuitive navigation structure that allows parents, teachers, staff, and public visitors to locate information with minimal effort.
  - **Consistent User Interface:** The website and application shall maintain consistent layouts, terminology, typography, icons, and navigation conventions to provide users with a familiar experience across the system.
  - **Accessible Information:** Information requested by the client, including the principal's opening message, vision and mission statements, staff information, school information, learner achievements, announcements, newsletters, homework, and learning resources, shall be presented in a clear and easily understandable manner.
  - **Responsive Website**: The website shall be responsive and adapt its layout to different screen sizes, including desktop computers, tablets, and mobile devices, to ensure that users can access school information regardless of their device.
- **Performance** (Satzinger et al, 2016)
  - **Website Loading Performance:** The website shall be optimised so that its pages load efficiently under normal internet conditions. Images used for the gallery and other visual content shall be appropriately compressed to reduce unnecessary loading times and data usage.
  - **Application Responsiveness:** The mobile application shall respond to normal user interactions without noticeable delays under normal operating conditions. Navigation between screens and access to previously synchronised information should be responsive.
- **Availability & Reliability** (Satzinger et al, 2016)
  - **Website Availability:** The school's website shall be hosted on a registered and accessible domain so that the school has a permanent online presence. The website shall not depend on the previous website or its unknown domain, as requested by the client.
  - **Application Reliability:** The mobile application shall provide reliable access to school information and communication features. Appropriate error handling shall be implemented to prevent application failures from unnecessarily interrupting the user's experience.

- **Data Synchronisation**
  - Where the application supports offline functionality, information accessed offline shall be synchronised with the central system when an internet connection becomes available.
- **Security & Privacy**(Satzinger et al, 2016)
  - **Role-Based Access Control:** The application shall implement role-based access so that parents, teachers, and administrators can only access information and functionality appropriate to their roles.
  - **Protection of Personal Information:** The system shall be designed with the Protection of Personal Information Act (POPIA) in mind. Personal information relating to learners, parents, teachers, and staff shall only be collected, processed, stored, and displayed for legitimate purposes and to authorised users.
  - **Secure Authentication:** Authenticated areas of the application shall require secure user authentication. Passwords and other authentication information shall not be stored or transmitted in an insecure manner.
- **Maintainability** (Satzinger et al, 2016)
  - **Maintainable Content:** The system shall allow authorised school personnel to update information without requiring major changes to the underlying application or website.
  - **Scalable Architecture:** The proposed system shall be designed in a manner that allows additional functionality, users, content, and future school requirements to be incorporated without requiring a complete redevelopment of the system.
- **Content & Media Requirements**
  - **Image Quality and Optimisation:** Images displayed on the website and application, including staff photographs, gallery images, and learner achievement content, shall maintain an appropriate level of quality while being optimised to avoid excessive storage requirements and loading times.
  - **Content Consistency:** Information presented on the website and mobile application shall be accurate, consistent, and kept up to date by authorised personnel.
  - **Content Management:** The system shall support the addition, modification, and removal of relevant school content, including newsletters, teacher announcements, homework, learning resources, and staff information, by authorised users.

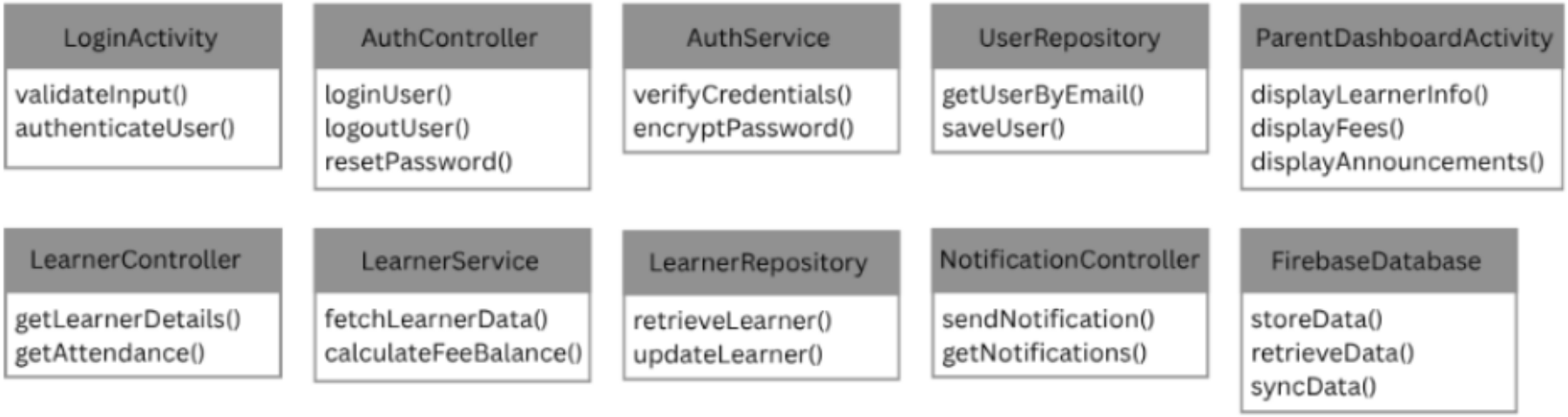
The non-functional requirements were primarily informed by the client's operational expectations and the school's existing circumstances. In particular, the client's requirement for the website to be hosted under a new, accessible domain addresses the loss of access to the previous website's domain. The requested website content and application features also highlighted the need for a system that is easy to maintain, reliable, accessible across devices, secure, and capable of supporting regularly changing school information. Furthermore, the application's intended use within the school's community supports the inclusion of data-conscious and offline capabilities, while the handling of learner, parent, and staff information necessitates appropriate security and POPIA considerations.

ANALYSIS ARTIFACTS

DOMAIN CLASS DIAGRAM (SATZINGER ET AL, 2016)



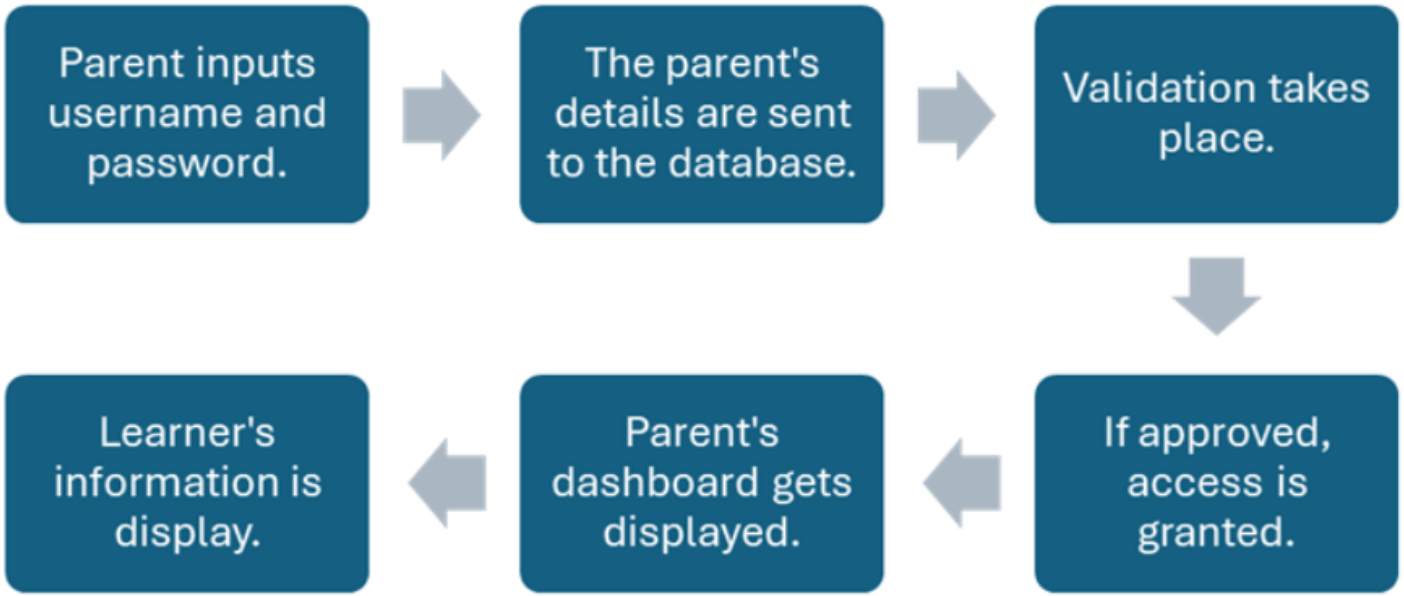
DESIGN CLASS DIAGRAM (SATZINGER ET AL, 2016)



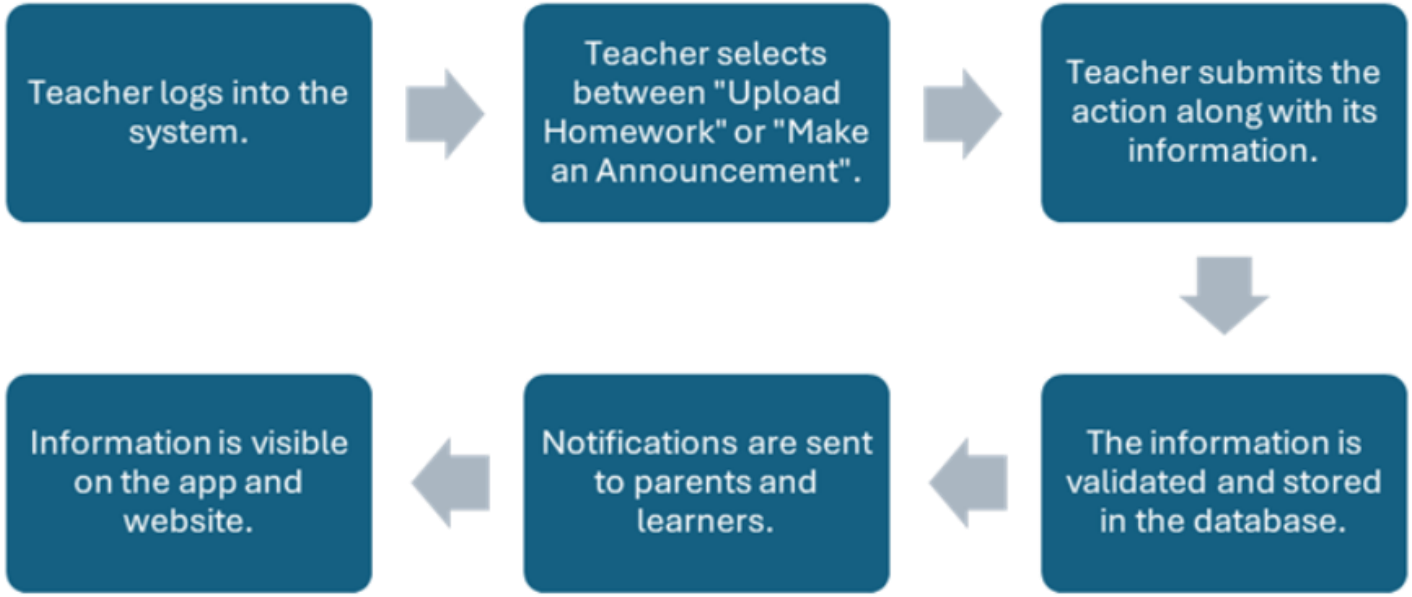
IMPLEMENTATION

SEQUENCE DIAGRAMS (SATZINGER ET AL, 2016)

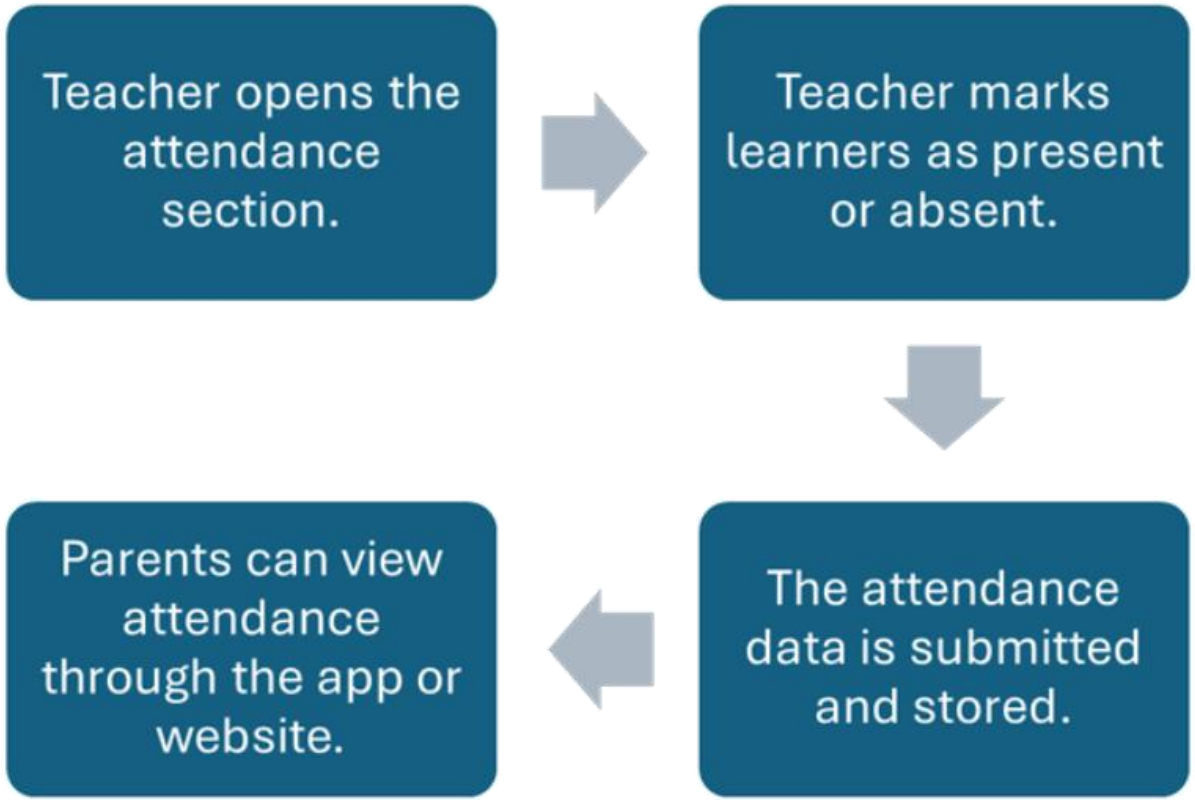
Parent login and view learner information sequence flow:



Teacher uploading homework and/or announcements sequence flow:

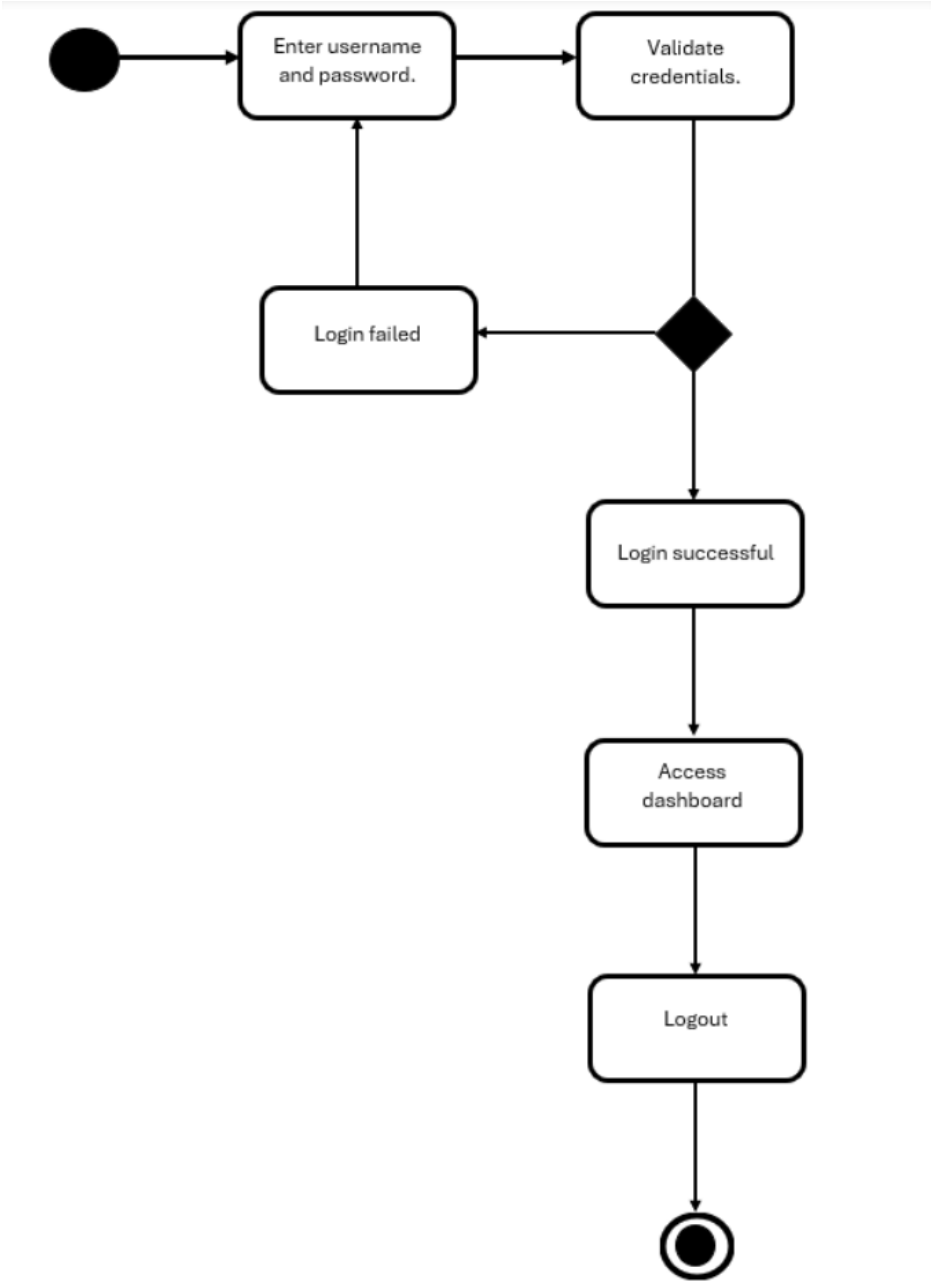


Learner attendance sequence flow:

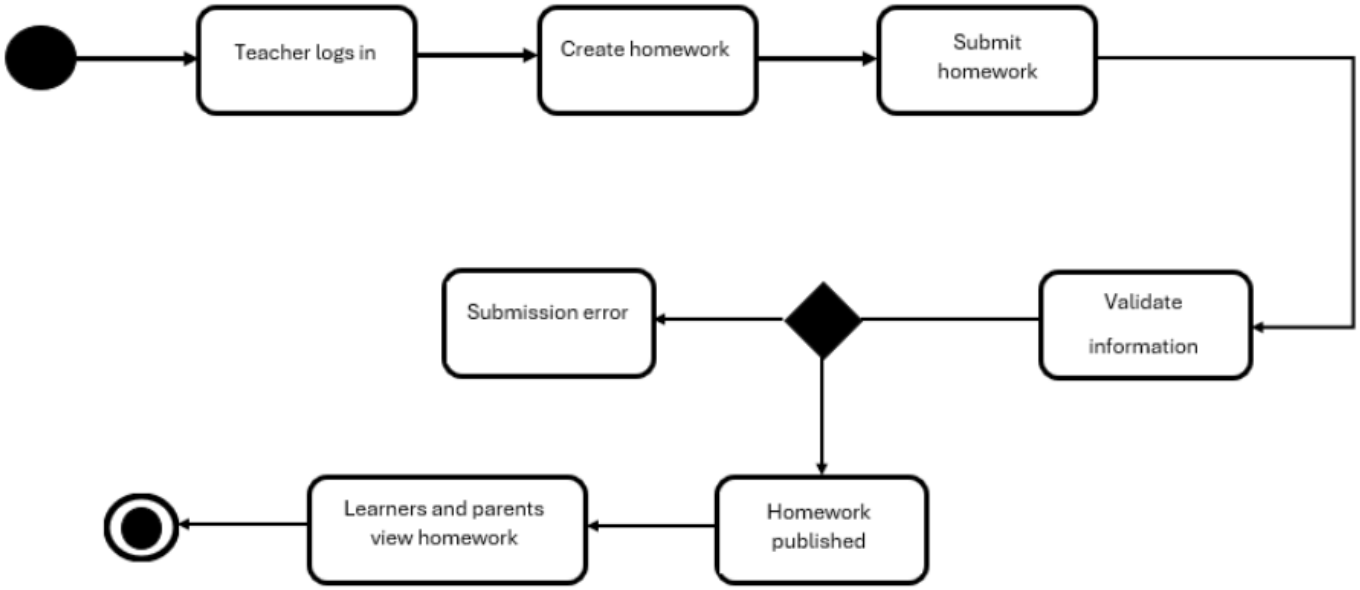


STATE DIAGRAMS (SATZINGER ET AL, 2016)

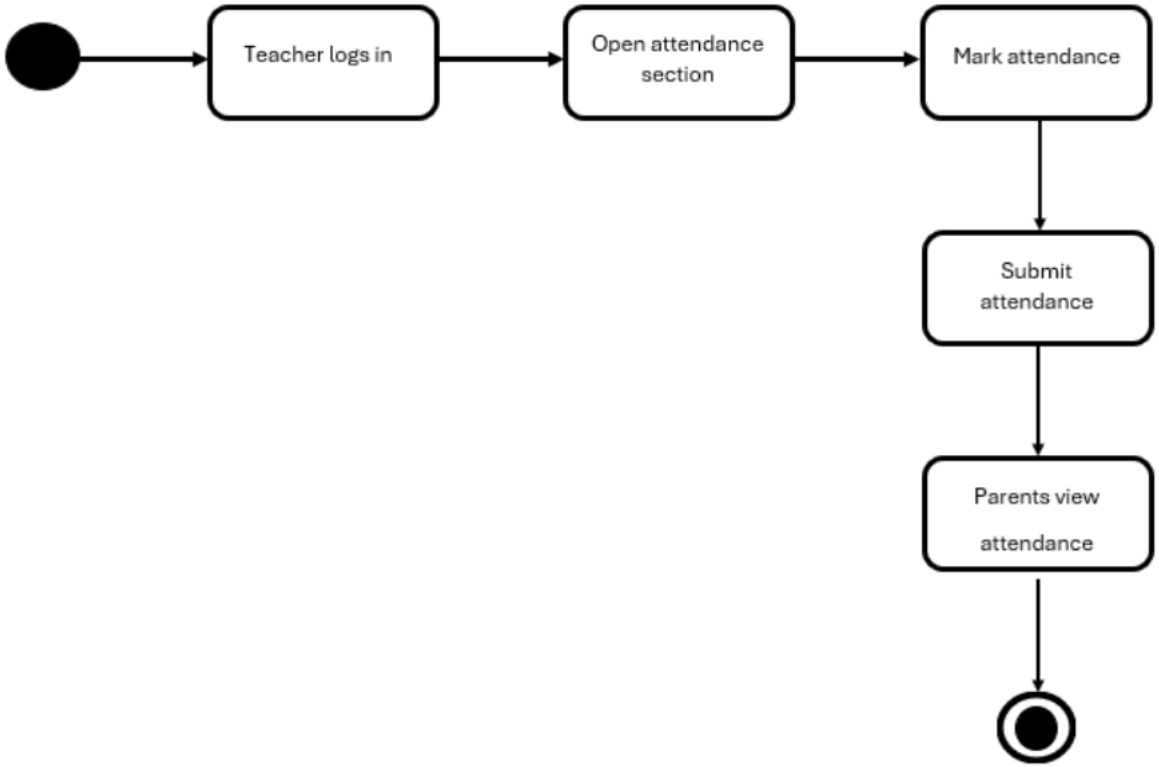
User authentication state diagram:



Homework uploading process state diagram:



Attendance state diagram:



DEPLOYMENT PLAN

The deployment plan defines how the Koos Matli Primary School website and mobile application will move from development through testing and into the production environment. The deployment process will focus on ensuring that the system is secure, reliable, maintainable and accessible to its different users.

Development Environment

The system will first be developed in a controlled development environment where each team member can develop and test their assigned components before they are integrated into the complete system:

- .NET MAUI will be used to develop the cross-platform mobile application for parents, teachers and administrators.~~Android Studio will be used for development of the Android mobile application.~~
- ASP.NET MVC will be used to develop the school website, providing public access to school information, announcements, admissions and other relevant content.
- Node.js will be used to develop the backend services and REST APIs responsible for processing application requests and managing communication between the mobile application, website and supporting services.~~ASP.NET Core will be used to develop the backend services and APIs.~~
- Firebase will be used to provide cloud-based services, including authentication, data management and storage.~~A suitable database management system will be used to store user, learner, attendance, homework, announcement and other system information.~~
- Git/GitHub will be used for version control, allowing team members to collaborate, track changes and maintain previous versions of the system.
- Separate development configurations will be used so that testing data and credentials are not exposed in the production environment.

Testing Environment

Before deployment, the system will be placed in a testing environment that closely represents the intended production environment (Satzinger et al, 2016). Testing will be performed to identify errors and confirm that the system meets the client's requirements. Testing will include:

- **Unit testing**: Testing individual components and functions.
- **System testing**: Testing the complete website and mobile application.
- **Integration testing**: Ensuring that the front end, backend, database and APIs communicate correctly.

- **Usability testing**: Ensuring that parents, teachers and administrators can navigate the system effectively.
- **Security testing**: Checking authentication, role-based access and protection of user information.
- **Responsive testing**: Ensuring that the website works correctly across different screen sizes and devices.

Testing will verify important functionality such as login and registration, role-based dashboards, notifications, timetable information, announcements, homework, applications and secure storage of information. Any defects identified during testing will be documented, corrected and retested before the system is approved for production deployment environment (Satzinger et al, 2016).

Production Deployment

Once testing has been successfully completed, the website and backend services will be deployed to a cloud hosting environment. The production environment will include:

- A publicly accessible school website.
- A secure HTTPS connection to protect information transmitted between users and the system.
- Deployment of the Node.js backend services and REST APIs.~~ASP.NET Core backend and APIs.~~
- Connection to the production database.
- Secure configuration of authentication and application settings.
- Regular monitoring of the system to identify errors or service interruptions.
- Backup and recovery procedures to reduce the risk of data loss.

The .NET MAUI mobile application will be built as a release version and packaged as an APK for distribution to authorised parents, teachers and administrators. The application will communicate with the production Node.js backend services and Firebase cloud services to securely access and manage system data.~~The Android application will be built as a release version and packaged as an APK for distribution to authorised parents, teachers and administrators. The application will communicate with the same production backend and database used by the system.~~

Database Deployment

The production database will contain the information required for the operation of the school system, including:

- Learner records.
- Parent and teacher accounts.

- Staff information.
- Attendance records.
- Homework and learning resources.
- Timetables and school events.
- Applications and application statuses.
- School announcements and notifications.

Database deployment will include creating the required tables, relationships and constraints before the production system is connected to the database. Database changes will be controlled through appropriate migration/versioning procedures to prevent accidental loss or corruption of information. The database will also have:

- Regular automated backups.
- Secure access controls.
- Protection against unauthorised access.
- A recovery procedure for restoring data following a failure.
- Appropriate separation between development/testing data and production data.

Release, Recovery & Maintenance

The deployment process will use version control to ensure that only approved versions of the website, application and backend are released. Before every production release:

- The updated system will be tested in the testing environment.
- The database changes will be reviewed.
- A backup of the production database will be performed.
- The approved version will be deployed.
- The system will be checked after deployment to ensure that the main functionality is operational.

If a serious problem occurs after deployment, the team will be able to roll back to the previous stable version and restore the database from a recent backup where necessary. After deployment, the system will be maintained through bug fixes, security updates, performance monitoring and future improvements based on feedback from the school.

DATA SCHEMA

CHOICE OF TECHNOLOGY

The proposed Koos Matli Primary School system will use a Relational Database Management System (RDBMS) as the primary data-storage technology. A relational database is appropriate because the system will manage structured information with clear relationships between entities such as learners, parents, teachers, classes, subjects, timetables, applications, attendance, homework, announcements, events, and school information (IIE, 2025). Storing these entities in related tables will allow the system to organise information efficiently while reducing unnecessary duplication and maintaining data consistency.

The database will contain separate tables for the major entities within the system. For example, a Learner table will contain learner information, while Parent/Guardian, Teacher, and Administrator tables will contain the relevant user information and roles. Additional tables will store classes, subjects, timetables, attendance records, homework and learning resources, announcements, newsletters, events, school applications, fees, and application statuses. Relationships between these tables will allow information to be associated with the correct users and learners. For example, a parent can be associated with their own child or children, while a teacher can be associated with the classes and subjects they teach.

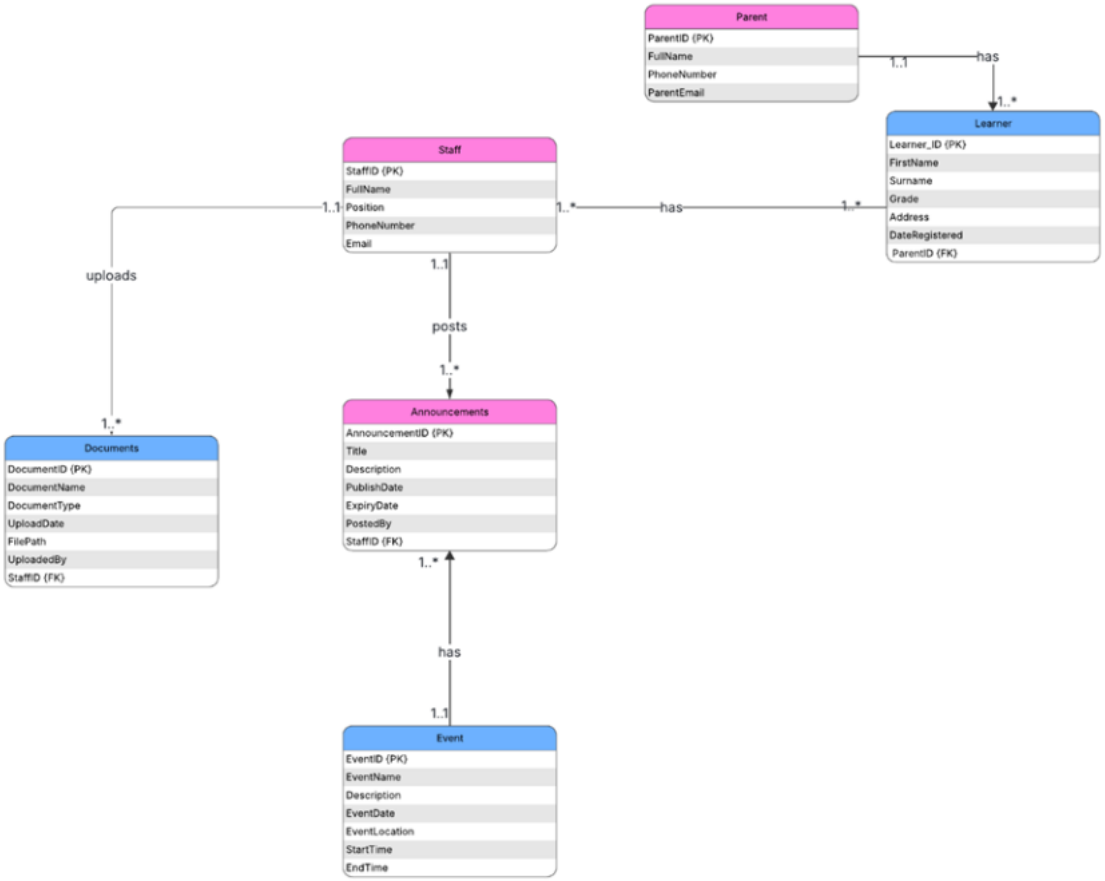
The relational database will also support data integrity and consistency through the use of primary keys, foreign keys, appropriate data types, validation rules, and constraints (IIE, 2025). This is particularly important because the system will manage learner and parent information that must be accurately associated with the correct accounts. Role-based access will be implemented at the application and backend levels so that users only receive access to information appropriate to their role. For example, a parent will only be permitted to access their own children's information, while teachers and administrators will receive different levels of access based on their responsibilities.

The database will be accessed through the Node.js backend and REST APIs, which will act as the intermediary between the database, the ASP.NET MVC website and the .NET MAUI mobile application. Firebase will provide the cloud-based services required for data management, authentication and storage.~~The database will be accessed through the ASP.NET Core backend/API, which will act as the intermediary between the database and the website or mobile application~~. Users will therefore not communicate directly with the database. This architecture provides an additional layer through which authentication, authorisation, validation, and security controls can be applied before information is retrieved or modified.

The database will be deployed within the system's cloud environment and will incorporate regular backups and recovery procedures to reduce the risk of permanent data loss. Development and testing databases will be separated from the production database to prevent test data or development changes from affecting the school's live information. Database migrations and version control will also be used to manage structural changes as the system develops.

A relational database was therefore selected because it provides the system with structured data storage, strong relationships between entities, data integrity, security, maintainability, and the ability to accommodate future growth. This makes it suitable for supporting both the school's public website and the role-based administration and communication functionality of the mobile application.

ENTITY-RELATIONSHIP DIAGRAM (IIE, 2025)


Parent
ParentID (PK)
FullName
PhoneNumber
ParentEmail
1..1
has
1..*
Learner
Learner_ID (PK)
FirstName
Surname
Grade
Address
DateRegistered
ParentID (FK)
Staff
StaffID (PK)
FullName
Position
PhoneNumber
Email
1..1
1..*
has
1..*
uploads
1..1
posts
1..*
1..*
Documents
DocumentID (PK)
DocumentName
DocumentType
UploadDate
FilePath
UploadedBy
StaffID (FK)
Announcements
AnnouncementID (PK)
Title
Description
PublishDate
ExpiryDate
PostedBy
StaffID (FK)
1..*
has
1..1
Event
EventID (PK)
EventName
Description
EventDate
EventLocation
StartTime
EndTime

The Entity Relationship Diagram (ERD) illustrates how the main data entities within the Koos Matli Primary School system are structured and how they relate to one another. The database uses a relational structure, where each entity represents a table containing related information, while primary keys (PKs) uniquely identify records and foreign keys (FKs) establish relationships between tables. The main entities in the system are Parent, Learner, Staff, Announcements, Documents, and Event.

Parent and Learner

The Parent entity stores information about each parent or guardian, including the ParentID, full name, phone number and email address. A parent can be associated with one or more learners, allowing the system to support parents who have multiple children attending the school. The Learner entity stores information such as the learner's ID, name, surname, grade, address and registration date. The ParentID (FK) connects each learner to their parent or guardian.

Staff and Learner

The Staff entity stores information about school employees, including their StaffID, full name, position, phone number and email address. Staff members are connected to learners within the system, allowing staff to manage or interact with learner-related information. This relationship supports the administration and management of learners by authorised school staff.

Staff and Announcements

The Staff entity is also connected to Announcements. Staff members can create and publish announcements for the school community. The Announcements table contains fields such as AnnouncementID, title, description, publish date, expiry date and PostedBy, which identifies the staff member responsible for the announcement. This allows the system to keep track of who created each announcement.

Staff and Documents

Staff members can also upload documents to the system. The Documents entity stores information such as the document ID, name, type, upload date, file path and the staff member who uploaded it. The StaffID (FK) establishes the connection between the document and the staff member who uploaded it. This provides accountability and allows authorised users to identify the source of uploaded documents.

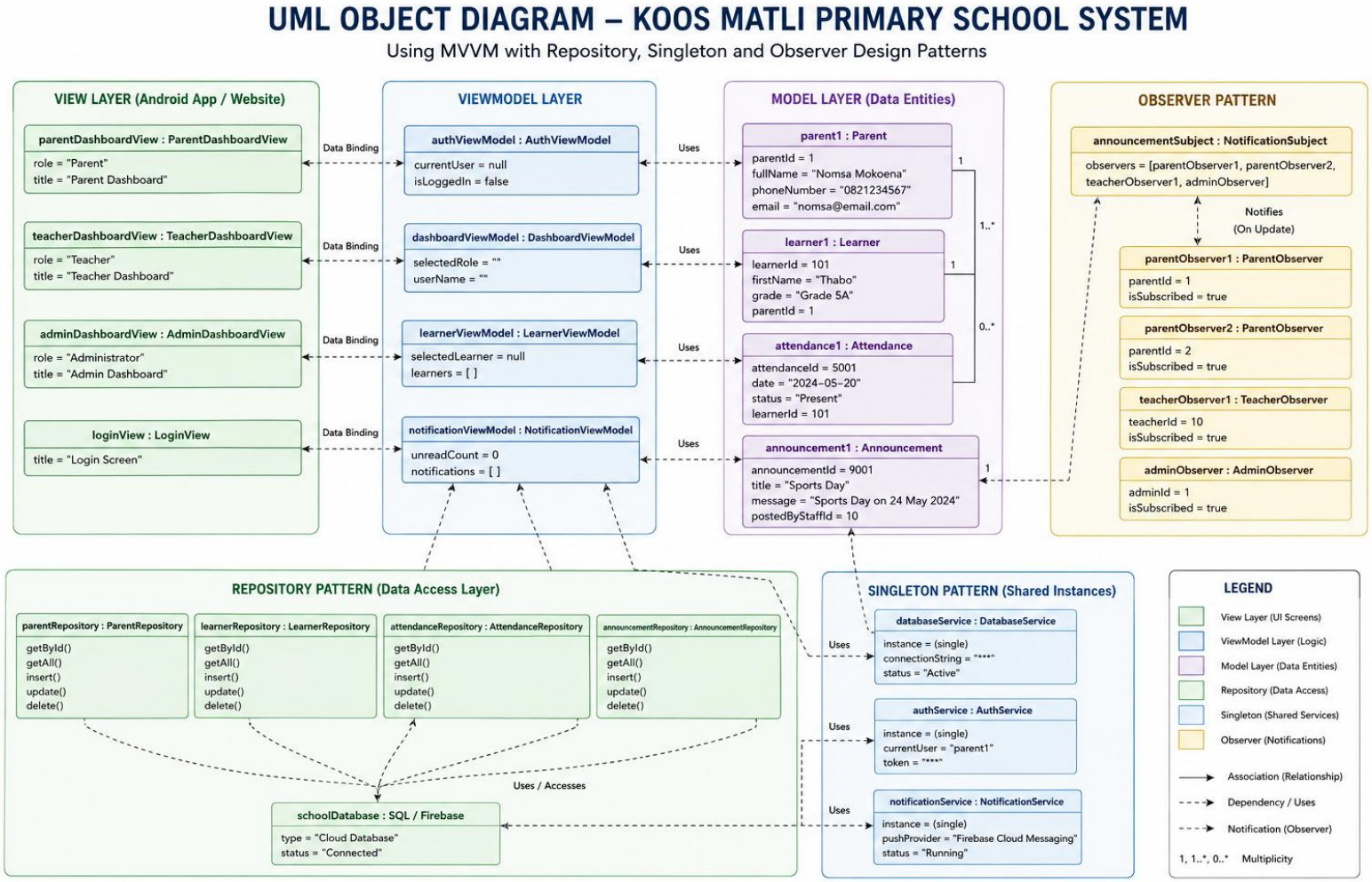
Announcements and Events

The Announcements entity is connected to the Event entity. Events contain information such as the event ID, event name, description, event date, location, start time and end time. This relationship allows school events to be associated with announcements so that important events can be communicated to parents and other users through the system.

Overall, the ERD provides a structured way of managing the school's information. Parents are linked to learners, staff manage learners and publish announcements, staff can upload documents, and announcements can be associated with school events. The use of primary and foreign keys helps maintain data integrity and ensures that information can be connected across the system without unnecessary duplication. This database structure supports the core functionality of the Koos Matli Primary School website and mobile application, while also providing a foundation for secure and organised data storage.

ARCHITECTURE ARTIFACTS

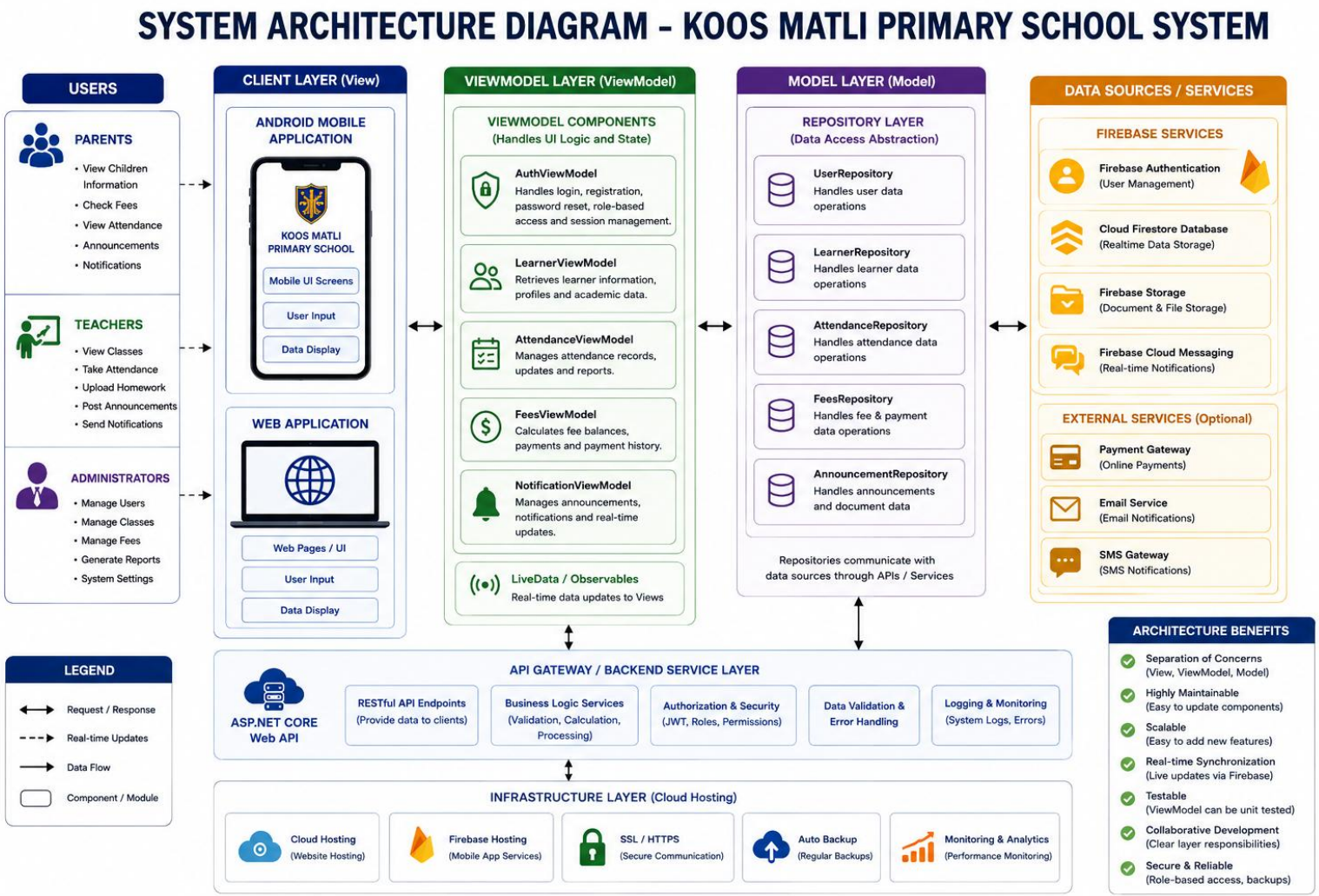
DESIGN PATTERNS (IIE, 2026)



The most suitable design pattern for the Koos Matli Primary School system is the Model-View-ViewModel (MVVM) design pattern together with supporting patterns such as Repository, Singleton, and Observer. MVVM is highly effective for mobile applications because it separates the user interface from the business logic and data handling processes (IIE, 2026). In this system, the View layer would represent the Android application screens and website pages that parents, teachers, and administrators interact with. The ViewModel layer would manage application logic such as login validation, fee calculations, learner data processing, and notification handling. The Model layer would manage the actual data entities such as learners, parents, attendance records, and announcements stored within the database. This separation improves maintainability, simplifies debugging, and allows developers to update features without affecting the entire system.

The Repository Pattern is also highly beneficial for this project because it creates a clear separation between the application logic and the database operations (IIE, 2026). Instead of allowing the application to communicate directly with Firebase or SQL databases, repository classes would manage all data retrieval and storage functions. This approach improves scalability and allows the database technology to be changed in the future with minimal effect on the rest of the application. The Singleton Pattern is appropriate for components such as database connections, authentication services, and notification managers because only one shared instance should exist throughout the application lifecycle (IIE, 2026). This prevents duplicate connections and improves performance efficiency.

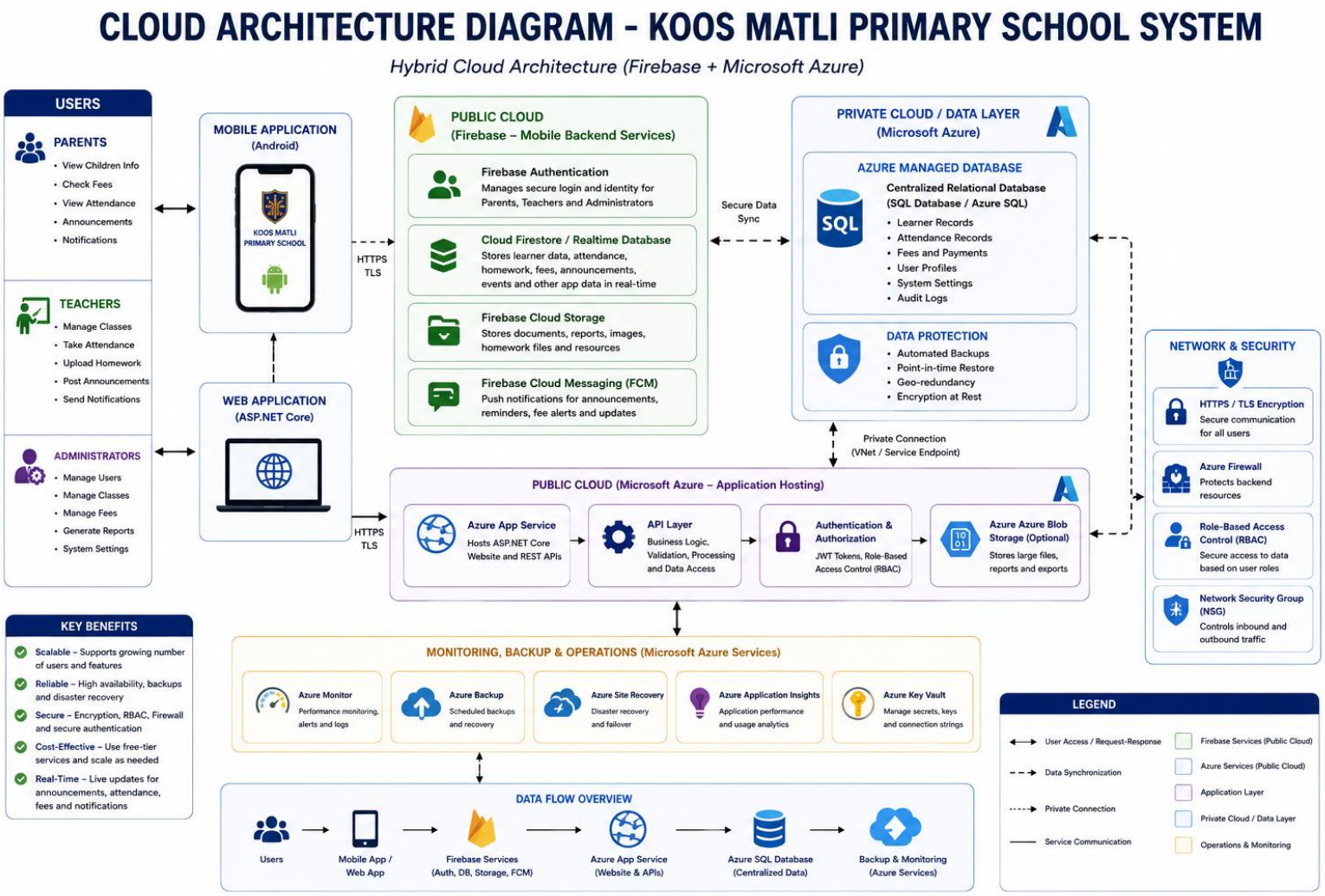
Additionally, the Observer Pattern is highly suitable because the system requires real-time communication and notifications between the school and parents. For example, when teachers upload announcements, homework, attendance records, or fee reminders, parents should automatically receive updates through the mobile application. The Observer Pattern supports this event-driven communication structure and improves responsiveness within the platform. Overall, these design patterns provide strong modularity, flexibility, maintainability, and scalability, making them highly appropriate for the school management system.


SYSTEM ARCHITECTURE DIAGRAM – KOOS MATLI PRIMARY SCHOOL SYSTEM
USERS
PARENTS
• View Children Information
• Check Fees
• View Attendance
• Announcements
• Notifications
TEACHERS
• View Classes
• Take Attendance
• Upload Homework
• Post Announcements
• Send Notifications
ADMINISTRATORS
• Manage Users
• Manage Classes
• Manage Fees
• Generate Reports
• System Settings
CLIENT LAYER (View)
ANDROID MOBILE APPLICATION
KOOS MATLI PRIMARY SCHOOL
Mobile UI Screens
User Input
Data Display
WEB APPLICATION
Web Pages / UI
User Input
Data Display
VIEWMODEL LAYER (ViewModel)
VIEWMODEL COMPONENTS (Handles UI Logic and State)
AuthViewModel
Handles login, registration, password reset, role-based access and session management.
LearnerViewModel
Retrieves learner information, profiles and academic data.
AttendanceViewModel
Manages attendance records, updates and reports.
FeesViewModel
Calculates fee balances, payments and payment history.
NotificationViewModel
Manages announcements, notifications and real-time updates.
LiveData / Observables
Real-time data updates to Views
MODEL LAYER (Model)
REPOSITORY LAYER (Data Access Abstraction)
UserRepository
Handles user data operations
LearnerRepository
Handles learner data operations
AttendanceRepository
Handles attendance data operations
FeesRepository
Handles fee & payment data operations
AnnouncementRepository
Handles announcements and document data
Repositories communicate with data sources through APIs / Services
DATA SOURCES / SERVICES
FIREBASE SERVICES
Firebase Authentication
(User Management)
Cloud Firestore Database
(Realtime Data Storage)
Firebase Storage
(Document & File Storage)
Firebase Cloud Messaging
(Real-time Notifications)
EXTERNAL SERVICES (Optional)
Payment Gateway
(Online Payments)
Email Service
(Email Notifications)
SMS Gateway
(SMS Notifications)
LEGEND
Request / Response
Real-time Updates
Data Flow
Component / Module
API GATEWAY / BACKEND SERVICE LAYER
ASP.NET CORE Web API
RESTful API Endpoints
(Provide data to clients)
Business Logic Services
(Validation, Calculation, Processing)
Authorization & Security
(JWT, Roles, Permissions)
Data Validation & Error Handling
Logging & Monitoring
(System Logs, Errors)
INFRASTRUCTURE LAYER (Cloud Hosting)
Cloud Hosting
(Website Hosting)
Firebase Hosting
(Mobile App Services)
SSL / HTTPS
(Secure Communication)
Auto Backup
(Regular Backups)
Monitoring & Analytics
(Performance Monitoring)
ARCHITECTURE BENEFITS
Separation of Concerns
(View, ViewModel, Model)
Highly Maintainable
(Easy to update components)
Scalable
(Easy to add new features)
Real-time Synchronization
(Live updates via Firebase)
Testable
(ViewModel can be unit tested)
Collaborative Development
(Clear layer responsibilities)
Secure & Reliable
(Role-based access, backups)

The most appropriate system architecture for the Koos Matli Primary School platform is the Model-View-ViewModel (MVVM) architecture. This architecture is especially suitable for Android mobile application development because it separates the presentation layer, business logic, and data access layers into independent components. The View layer is responsible for displaying information to the user through mobile application screens and website interfaces. The ViewModel acts as the intermediary between the View and the Model by handling logic such as authentication, fee calculations, learner data retrieval, and notification management. The Model layer stores and manages the actual data structures and communicates with the database through repositories and APIs (IIE, 2026).

MVVM architecture provides several advantages for this project. Firstly, it improves maintainability because developers can update one layer without significantly affecting the others. Secondly, it enhances scalability since new features such as report cards, online payments, or additional communication tools can be added more easily in the future. Thirdly, MVVM supports real-time data synchronization very effectively, especially when integrated with Firebase services. Since the application requires live updates for announcements, attendance records, fee balances, and notifications, MVVM works well with asynchronous data flows and LiveData components. Furthermore, the architecture improves testability because business logic within the ViewModel can be independently unit tested without relying on the user interface. This leads to more stable and reliable software development.

The architecture also supports better collaboration among developers because responsibilities are clearly separated. Front-end developers can focus on designing user interfaces while backend developers manage business logic and database communication independently (IIE, 2026). Overall, MVVM provides a clean, organized, scalable, and maintainable architecture that aligns well with the functional and non-functional requirements of the school platform.


CLOUD ARCHITECTURE DIAGRAM - KOOS MATLI PRIMARY SCHOOL SYSTEM
Hybrid Cloud Architecture (Firebase + Microsoft Azure)
USERS
PARENTS
• View Children Info
• Check Fees
• View Attendance
• Announcements
• Notifications
TEACHERS
• Manage Classes
• Take Attendance
• Upload Homework
• Post Announcements
• Send Notifications
ADMINISTRATORS
• Manage Users
• Manage Classes
• Manage Fees
• Generate Reports
• System Settings
MOBILE APPLICATION (Android)
KOOS MATLI PRIMARY SCHOOL
WEB APPLICATION (ASP.NET Core)
HTTPS TLS
HTTPS TLS
PUBLIC CLOUD (Firebase – Mobile Backend Services)
Firebase Authentication
Manages secure login and identity for Parents, Teachers and Administrators
Cloud Firestore / Realtime Database
Stores learner data, attendance, homework, fees, announcements, events and other app data in real-time
Firebase Cloud Storage
Stores documents, reports, images, homework files and resources
Firebase Cloud Messaging (FCM)
Push notifications for announcements, reminders, fee alerts and updates
Secure Data Sync
PRIVATE CLOUD / DATA LAYER (Microsoft Azure)
AZURE MANAGED DATABASE
SQL
Centralized Relational Database (SQL Database / Azure SQL)
• Learner Records
• Attendance Records
• Fees and Payments
• User Profiles
• System Settings
• Audit Logs
DATA PROTECTION
• Automated Backups
• Point-in-time Restore
• Geo-redundancy
• Encryption at Rest
Private Connection (VNet / Service Endpoint)
PUBLIC CLOUD (Microsoft Azure – Application Hosting)
Azure App Service
Hosts ASP.NET Core Website and REST APIs
API Layer
Business Logic, Validation, Processing and Data Access
Authentication & Authorization
JWT Tokens, Role-Based Access Control (RBAC)
Azure Azure Blob Storage (Optional)
Stores large files, reports and exports
NETWORK & SECURITY
HTTPS / TLS Encryption
Secure communication for all users
Azure Firewall
Protects backend resources
Role-Based Access Control (RBAC)
Secure access to data based on user roles
Network Security Group (NSG)
Controls inbound and outbound traffic
KEY BENEFITS
Scalable – Supports growing number of users and features
Reliable – High availability, backups and disaster recovery
Secure – Encryption, RBAC, Firewall and secure authentication
Cost-Effective – Use free-tier services and scale as needed
Real-Time – Live updates for announcements, attendance, fees and notifications
MONITORING, BACKUP & OPERATIONS (Microsoft Azure Services)
Azure Monitor
Performance monitoring, alerts and logs
Azure Backup
Scheduled backups and recovery
Azure Site Recovery
Disaster recovery and failover
Azure Application Insights
Application performance and usage analytics
Azure Key Vault
Manage secrets, keys and connection strings
DATA FLOW OVERVIEW
Users
Mobile App / Web App
Firebase Services (Auth, DB, Storage, FCM)
Azure App Service (Website & APIs)
Azure SQL Database (Centralized Data)
Backup & Monitoring (Azure Services)
LEGEND
User Access / Request-Response
Data Synchronization
Private Connection
Service Communication
Firebase Services (Public Cloud)
Azure Services (Public Cloud)
Application Layer
Private Cloud / Data Layer
Operations & Monitoring

The most suitable cloud architecture for the Koos Matli Primary School system is Hybrid Cloud Architecture. A hybrid cloud combines public cloud services with secure centralized data management to provide scalability, reliability, and cost-effectiveness (IIE, 2026). This approach is highly suitable because the proposed solution includes both a mobile application and a website that must share the same centralized database, authentication services, notifications, and cloud storage resources. Using a hybrid cloud environment allows the school to maintain secure access to sensitive learner and parent data while also benefiting from scalable cloud infrastructure.

Firebase services would be ideal for the mobile backend because they provide real-time database synchronization, authentication services, cloud messaging, and cloud storage. Firebase Authentication would securely manage parent, teacher, and administrator login credentials, while Firebase Realtime Database or Firestore would store learner records, attendance, homework, fee balances, and announcements. Firebase Cloud Messaging would enable push notifications for reminders and announcements. The website component could be hosted using ASP.NET Core on Microsoft Azure, which offers secure web hosting, HTTPS support, monitoring tools, backup systems, and scalability features.

Hybrid cloud architecture offers numerous benefits for the project. It improves scalability by allowing the system to support increasing numbers of learners, parents, and future system features without requiring major infrastructure changes. It also improves reliability because cloud platforms provide automated backups, disaster recovery solutions, and high uptime availability (IIE, 2026). Security is enhanced through HTTPS encryption, role-based access control, secure authentication, firewall protection, and encrypted database communication. Additionally, hybrid cloud services are cost-effective because the school can initially use free-tier cloud services and later upgrade as the number of users increases. Overall, the hybrid cloud architecture provides a secure, scalable, reliable, and future-ready solution for the school management system.

SECURITY

Security is a critical requirement for the Koos Matli Primary School system because the platform will process and store sensitive information relating to learners, parents/guardians, teachers, administrators, school documents, attendance, homework and announcements. Security will therefore be incorporated throughout the system development lifecycle rather than only being addressed after an attack has occurred. The system will follow a defence-in-depth approach, where multiple security controls are implemented at the application, database, network and user levels. The main objectives are to protect the confidentiality, integrity and availability of school information while ensuring that users only have access to information and functions appropriate to their roles.

Security Analysis

The main information that requires protection includes:

- Learner personal information and school records.
- Parent/guardian contact information.
- Staff and administrator account details.
- Attendance and homework records.
- School announcements and events.
- Uploaded school documents and resources.
- Authentication credentials.
- System and administrative information.

The system will use security controls to reduce the likelihood and impact of threats such as unauthorised access, data breaches, weak passwords, malware, data loss, interception of information, insecure input, and misuse of user privileges (Ciampa, 2022).

Security Risks and Mitigation

Unauthorised Access

- Risk: An unauthorised person could gain access to the staff or administrative sections of the system and view or modify sensitive school information.
- Mitigation:
    - User authentication: Users will be required to log in using valid credentials before accessing protected features.
    - Role-Based Access Control (RBAC): Permissions will be assigned according to roles such as Parent, Teacher and Administrator.
    - Principle of least privilege: Users will only receive the minimum permissions required to perform their responsibilities.
    - Admin-only functionality: Sensitive functions such as managing users, approving applications and modifying system settings will only be available to authorised administrators.
    - Session management: Sessions will expire after a period of inactivity to reduce the risk of an unattended device being misused.
    - Multi-factor authentication: Where practical, additional verification can be introduced for administrator and other privileged accounts.

For example, a teacher may be allowed to upload homework and announcements but should not be able to modify administrator accounts or access information belonging to unrelated learners.

Weak or Compromised Passwords

- Risk: Weak, reused or compromised passwords could allow attackers to gain access to user accounts.
- Mitigation:
    - Passwords will have minimum complexity requirements.
    - Passwords will never be stored as plain text.
    - Passwords will be securely hashed using an appropriate password-hashing algorithm.
    - Login attempts will be monitored and temporarily restricted after repeated unsuccessful attempts.
    - Users will be encouraged to use unique passwords.
    - Password-reset functionality will require verification of the user's identity.
    - Administrator accounts will receive stronger authentication requirements where possible.
    - Password security will therefore protect both normal user accounts and privileged accounts from common credential-based attacks.

Data Breach

- Risk: Sensitive learner, parent or staff information could be exposed through unauthorised database access, compromised accounts or application vulnerabilities.
- Mitigation:
    - Encryption in transit: HTTPS/TLS will be used for communication between users, the website, APIs and cloud services.
    - Encryption at rest: Sensitive stored information and backups should be encrypted where supported by the chosen cloud/database platform.
    - Restricted database access: Users will not directly access the database. Database communication will occur through authorised application services and APIs.
    - Role-based permissions: Access to sensitive records will be restricted according to user roles.
    - Secure cloud configuration: Cloud resources will be configured using appropriate access controls and security settings.
    - Data minimisation: The system will only collect information that is necessary for the school's functions.
    - Secure document storage: Uploaded documents will be stored in controlled cloud storage rather than being exposed through publicly accessible folders.

Insecure Input and Injection Attacks

- Risk: Attackers could enter malicious information into forms such as login, registration, announcements or document-upload fields in an attempt to manipulate the application or database.
- Mitigation:
    - All user input will be validated on the server side.
    - Input will be checked for correct formats, lengths and expected values.
    - Parameterised queries or secure ORM/database methods will be used instead of constructing database queries directly from user input.
    - Output will be encoded appropriately to reduce cross-site scripting (XSS) risks.
    - File uploads will be restricted according to permitted file types and sizes.
    - Uploaded files will be stored securely and should be scanned where appropriate.

This will reduce the risk of attacks such as SQL injection, Cross-Site Scripting (XSS) and malicious file uploads.

Interception of Data During Transmission

- Risk: Sensitive information such as login credentials, learner information and parent contact details could be intercepted while travelling between the user's device and the system.
- Mitigation:
    - HTTPS will be enforced across the website.
    - TLS encryption will protect information transmitted between clients and servers.
    - Secure API communication will be used between application components.
    - Insecure HTTP connections will be redirected to HTTPS where applicable.
    - Secure cookies/session mechanisms will be used for web authentication.
    - Sensitive information will not be transmitted through unsecured communication channels.
    - These measures help prevent attackers from reading or modifying information while it is being transferred.

Malware and Malicious Activity

- Risk: Malicious software or attackers could attempt to compromise the website, application or uploaded documents.
- Mitigation:
    - The application framework, libraries and dependencies will be regularly updated.
    - Security patches will be applied when vulnerabilities are identified.
    - A Web Application Firewall (WAF) or equivalent cloud security controls can be used to filter malicious traffic.
    - Uploaded files will be restricted by type and size.
    - Administrative actions will be recorded in audit logs.
    - Unnecessary services and ports will be disabled.
    - Cloud hosting security controls will be configured appropriately.
    - Developers will avoid using outdated or unsupported software components.

Data Loss and System Failure

- Risk: Hardware failure, software errors, accidental deletion, cyberattacks or cloud service problems could result in the loss of important school information.
- Mitigation:
    - Automated database backups will be scheduled.
    - Important backups will be stored separately from the primary system.
    - Backup data will be protected using appropriate access controls and encryption.
    - A documented disaster recovery procedure will be created.
    - Backups will be periodically tested to ensure that they can actually be restored.
    - Version control such as Git will be used for source-code management.
    - Previous stable versions of the application will be retained so that problematic releases can be rolled back.
    - The objective is to minimise data loss and reduce downtime following a system failure.

Availability and Denial-of-Service Protection

- Risk: Attackers could overwhelm the website or APIs with excessive requests, making the system unavailable to parents, teachers and administrators.
- Mitigation:
    - Cloud hosting will provide scalable infrastructure where required.
    - Rate limiting will be applied to sensitive APIs and authentication endpoints.
    - Firewall and WAF controls can filter suspicious traffic.
    - Monitoring will identify unusual increases in traffic.
    - Backups and recovery procedures will support restoration after major incidents.
    - Critical services will be monitored for availability.
    - This supports the availability component of the CIA security model.

Data Protection and Privacy

Because the system will process personal information relating to learners, parents and staff, privacy will be incorporated into the system design. The system will follow the principles of data minimisation, controlled access, secure processing and responsible storage. Appropriate measures will be implemented to align the system with applicable South African privacy requirements, including the Protection of Personal Information Act (POPIA). Security measures will include:

- Collecting only information required for school-related purposes.
- Restricting access to personal information based on user roles.
- Protecting personal information during transmission and storage.
- Avoiding unnecessary exposure of learner information.
- Providing appropriate account and data-management procedures.
- Securely deleting information when it is no longer required, where applicable.
- Protecting backups containing personal information.

This ensures that privacy is considered during the design of the system rather than being added afterwards (Ciampa, 2022).

Security Monitoring and Audit Logging

The system will maintain logs of important security and administrative activities. These may include:

- Successful and unsuccessful login attempts.
- Password-reset attempts.
- User account creation and modification.
- Changes to learner records.
- Announcement creation or modification.
- Document uploads and deletions.
- Administrative activities.
- Suspicious or repeated requests.

Logs will help administrators identify unusual behaviour, investigate security incidents and determine what occurred after a suspected attack. Access to logs will also be restricted because they may contain sensitive system information (Ciampa, 2022).

Secure Software Development

Security will be incorporated throughout the Software Development Life Cycle (SDLC). Developers will:

- Follow secure coding practices.
- Validate and sanitise user input.
- Avoid hardcoding passwords, API keys or other secrets in source code.
- Store sensitive configuration information securely.
- Keep dependencies and frameworks updated.
- Use version control to track changes.
- Perform code reviews where possible.
- Test authentication and authorisation functions.
- Correct identified vulnerabilities before deployment.

Security testing will therefore take place during development instead of waiting until the final deployment stage.

Security Testing

Before the system is deployed, security-related tests will be performed to identify vulnerabilities. Testing will include:

- **Authentication testing:** Ensuring that users cannot access protected areas without valid credentials (Ciampa, 2022).
- **Authorisation testing:** Checking that users cannot access functions belonging to other roles. For example, a parent should not be able to access administrator functions.
- **Input validation testing:** Testing forms and inputs to ensure malicious or invalid information is rejected (Ciampa, 2022).
- **Session testing:** Checking that inactive sessions expire correctly and that users cannot continue using protected pages after logging out.
- **Database security testing:** Checking that users cannot directly access or manipulate database records without appropriate permissions.
- **Vulnerability testing:** Identifying outdated dependencies, insecure configurations and other known vulnerabilities before production deployment.= (Ciampa, 2022).

Incident Response

If a security incident occurs, the school will follow a documented incident-response process:

1. Identify - Detect and confirm the security incident.
2. Contain - Restrict affected accounts, systems or services.
3. Investigate - Examine logs and determine what occurred.
4. Recover - Restore affected services and data from trusted backups where necessary.
5. Review - Identify the cause and implement additional controls to prevent recurrence.
6. Report - Escalate and communicate the incident to the appropriate responsible parties where required.

Having a response process ensures that security incidents can be handled systematically rather than reactively (Ciampa, 2022).

Security Principles Applied

The system will follow the CIA Triad:

- **Confidentiality:** Learner, parent and staff information will only be accessible to authorised users. Encryption and access controls will protect sensitive information (Ciampa, 2022).
- **Integrity:** Only authorised users will be allowed to create, modify or delete records, ensuring that school information cannot be changed without permission (Ciampa, 2022).
- **Availability:** Backups, monitoring, reliable cloud hosting and disaster recovery procedures will help ensure that the website and application remain available when required (Ciampa, 2022).

Security will be treated as a fundamental requirement of the Koos Matli Primary School system rather than an additional feature added after development. The system will combine authentication, role-based authorisation, encryption, secure password management, input validation, database protection, secure cloud configuration, backups, monitoring, audit logging, security testing and incident response to protect school information. By applying the principles of confidentiality, integrity and availability, together with privacy and least-privilege principles, the system will provide a more secure environment for learners, parents, teachers and administrators. This layered approach also allows the security controls to be strengthened as the platform grows and new features are introduced.

DEVOPS

DevOps practices will be incorporated into the development of the Koos Matli Primary School system to support continuous development, collaboration, testing, version control, and reliable delivery of both the mobile application and website. The team will apply DevOps principles throughout the software development lifecycle to ensure that changes can be developed and integrated in a controlled manner while maintaining the quality, stability, and security of the system (IIE, 2026).

Version Control and Collaboration

GitHub will be used as the team's central version control and collaboration platform. All project source code and development changes will be managed through GitHub, allowing team members to work on the system without overwriting each other's work. GitHub will maintain a history of code changes, making it possible to identify who made a change and when it was made. This will provide an additional backup of the project's source code and allow the team to restore previous versions when necessary.

Branching Strategy and Development Flow

The team will use a main branch as the stable version of the system, while separate feature branches will be created for individual development tasks. For example, a developer may create a feature branch to work on authentication, database functionality, or a particular user interface feature without directly changing the stable version.

Once a feature has been completed and tested, the changes will be reviewed before being merged into the main branch. This branching strategy will reduce conflicts between team members, prevent incomplete features from affecting the stable system, and provide a structured development workflow.

The planned development flow will follow the general process of:

Creating a feature branch → Developing the feature → Testing the changes → Reviewing the changes → Merging the completed feature into the main branch.

This process will allow team members to develop different components of the website and application simultaneously while maintaining control over the overall project.

Automated Testing and Quality Assurance

Automated testing will be integrated into the development process to identify errors before changes are considered ready for the main version of the system. Automated tests will be used to verify important functionality and improve the reliability of the application. Testing will focus on areas such as:

- Login and authentication.
- Database connectivity.
- Input validation.
- User interface functionality.
- Parent functionality.
- Teacher functionality.
- Administrator functionality.
- Other critical system features.

Automated testing will reduce the need to repeatedly perform the same checks manually whenever changes are made. When new functionality is developed or existing functionality is modified, automated tests will be used to determine whether the changes continue to work correctly. This will help identify defects earlier in the development lifecycle and reduce the possibility of introducing new errors into previously working functionality.

Continuous Integration and Code Quality

The use of GitHub and automated testing will support a consistent development and integration process. Changes will be tested before being incorporated into the main branch, helping the team maintain a stable version of the system.

The team will aim to identify and resolve issues during development rather than waiting until the final deployment stage. Automated testing and code reviews will also contribute to improved code quality by ensuring that functionality is evaluated throughout development instead of only at the end of the project.

Where appropriate, the team will use GitHub Actions or a similar CI process to automatically run the project's tests when changes are pushed or submitted for integration. This will provide an additional quality check before changes are merged into the stable branch.

Deployment and Maintenance

DevOps practices will also support the deployment and ongoing maintenance of the Koos Matli Primary School system (IIE, 2026). Maintaining the project in GitHub will give the team access to previous versions of the source code, making future maintenance and troubleshooting easier. If a change introduces an unexpected problem, the team will be able to identify the change through the version history and, where necessary, restore an earlier working version. The same development, testing, review and deployment process will be followed when future features or updates are introduced.

Benefits of DevOps for the Project

The implementation of DevOps practices provides several benefits to the Koos Matli Primary School system (IIE, 2026):

- Improved collaboration: GitHub allows all team members to contribute to the same project while tracking changes.
- Controlled development: Feature branches prevent unfinished work from affecting the stable version.
- Improved reliability: Automated testing helps identify defects before changes are integrated.
- Better code quality: Testing and code review encourage developers to produce more reliable code.
- Faster development: Team members can work on different features simultaneously.
- Easier maintenance: GitHub's version history makes it easier to identify and reverse problematic changes.
- Reduced deployment risk: Changes are tested before being incorporated into the stable system.
- Future scalability: The same branching, testing, and version-control processes can be maintained as the system grows.

Overall, DevOps will be addressed in the Koos Matli Primary School project through the planned use of GitHub for version control and collaboration, a structured branching strategy for development, and automated testing for quality assurance. These practices will allow the team to manage changes systematically, identify errors earlier, protect the stability of the main version, and maintain a reliable development workflow (IIE, 2026). As a result, the planned DevOps approach will contribute to making the system maintainable, collaborative, reliable, secure, and suitable for future development and deployment.

RUNNING COSTS

Running Cost	Purpose	Year 1 – Monthly Estimated Costs	Year 2 – Monthly Estimated Costs	Two-Year Growth Assumption
Azure App Service	Hosts the ASP.NET Core website/API in production	R212	R413	Upgrade from B1 to B2 as traffic increases
Firebase Firestore	Stores learner, parent, staff, announcements, attendance and other system data	R0 – R50	R50 – R150	Usage increases with users and database operations
Firebase Authentication	Secure login for parents, teachers and administrators	R0	R0	Up to 50,000 monthly active users can remain within the no-cost allowance on the applicable Blaze/Identity Platform setup.
Firebase Cloud Storage	Stores documents, images and other school files	R0 – R30	R30 – R100	Storage requirements increase as more documents and images are uploaded
Firebase Cloud Messaging (FCM)	Push notifications for announcements, reminders and updates	R0	R0	Firebase Cloud Messaging is currently provided at no cost.
Backups & Recovery	Protection of database and important system information	R25	R50	Increased backup storage as data grows

Domain Name	School website domain registration	R25	R30	Estimated monthly equivalent of annual domain registration
SSL/HTTPS	Secure communication between users and the website	R0	R0	Included through the cloud hosting configuration/managed certificate
Monitoring & Error Tracking	Monitoring availability, application errors and system performance	R0 – R25	R25 – R50	Increased monitoring requirements as system usage grows
GitHub	Source-code version control and repository management	R0	R0	GitHub Free is sufficient for the project's basic version-control requirements
Estimated Monthly Total		**R262 – R342**	**R598 – R793**	Cost increases primarily through hosting capacity, storage and database usage

User Scale	Expected Usage	Azure Hosting	Firebase Database & Storage	Authentication & Notifications	Estimated Monthly Infrastructure Cost
100 users	Small school deployment; low database activity	R212	R0	R0	**R250 – R300**
1,000 users	Regular parent, teacher and administrator activity	R212	R0 – R50	R0	**R300 – R400**
10,000 users	High activity; significantly more database reads/writes, storage and traffic	R413 – R826*	R100 – R250	R0 – R50	**R650 – R1,150**

CHANGE MANAGEMENT

Change management is important for ensuring that the Koos Matli Primary School system is successfully adopted by the school and its users. The introduction of the website and mobile application will change how information is communicated, how documents are accessed, and how certain administrative activities are performed. Therefore, the project will use a structured change management approach that prepares the school, staff, parents, and other users for the new system while reducing resistance and disruption during the transition.

Koos Matli Primary School will adopt the proposed system because it provides a more efficient and centralised way of managing school information and communicating with parents. The website will provide the school with a professional digital presence where information such as admissions, announcements, events, staff information, achievements, contact details, and important documents can be accessed easily. The mobile application will further support communication and administration by providing parents, teachers, and administrators with role-specific functionality.

The system can also reduce the school's reliance on paper-based communication and manual processes. Announcements can be published digitally, documents can be uploaded and accessed electronically, and parents can receive important information without always needing to visit the school. This can reduce printing requirements, improve the speed of communication, and make school information easier to manage.

The school will also benefit from having information stored in a more structured and centralised system. Role-based access will ensure that staff members only have access to the functions and information required for their responsibilities.

Organisational Adoption Strategy

The team will involve the school and relevant staff members throughout the implementation process rather than introducing the completed system without preparation. The principal and selected staff members will be consulted to confirm that the system meets the school's operational requirements. The adoption strategy will include:

- Stakeholder engagement: The principal and relevant staff members will be shown the system and its benefits before deployment.
- User involvement: Staff feedback will be collected during testing to identify problems and usability improvements.
- Demonstrations: The team will demonstrate the major website and application functions before the system goes live.
- Training: Staff members will receive practical training on the functions relevant to their roles.
- Documentation: User guides and system documentation will be provided for future reference.
- Gradual implementation: The system will initially be introduced in a controlled manner before becoming the school's primary digital platform.

- Post-deployment support: The team will monitor issues and provide assistance during the initial adoption period.
- User Adoption: How and Why Users Will Adopt the Software

The main users of the system will include parents, teachers, and school administrators. Users are expected to adopt the system because it provides convenient access to information and reduces the need to rely on paper-based communication or physical visits to the school.

Parents will be able to use the mobile application to access relevant school information, view their children's timetables, receive announcements, view events, access documents, and communicate with the school. The website will also allow parents and community members to access general school information without requiring them to contact the school directly.

Teachers will benefit from having a central platform through which they can manage relevant information, view their schedules and classes, communicate announcements, and perform their assigned functions. Administrators will benefit from improved management of users, applications, staff information, announcements, and other administrative activities.

User Adoption Strategy

To encourage adoption, users will be introduced to the system through the school's existing communication channels. Parents will be informed about the new system and provided with clear instructions on how to access the website and mobile application.

Training and demonstrations will focus on practical tasks rather than technical concepts. For example, parents will be shown how to log in, access announcements, view schedules, access documents, and manage their profiles. Teachers will be trained on functions such as managing announcements, viewing classes and schedules, and performing their assigned administrative activities. Administrators will receive more comprehensive training covering user management, applications, staff management, announcements, and system administration.

The system's simple interface, role-based navigation, offline-first approach, and data-conscious design will also support adoption by making the system easier and more practical for the school's community.

Transition from the Existing System to the New System

The transition will be managed carefully to minimise disruption to the school. The existing communication and administrative processes will initially continue while the new system is tested and prepared for deployment. This allows the team to verify that important functionality works correctly before the school becomes fully dependent on the new system.

	Stage	Activity	Description
1	Preparation	Identify existing processes	Document how the school currently manages announcements, documents, schedules, and communication.
2	System Setup	Configure production system	Deploy the website, application backend, database, authentication and required services.
3	Data Preparation	Prepare existing information	Identify information that needs to be transferred into the new system and ensure that it is accurate
4	User Training	Train staff and selected users	Demonstrate the system and provide user guides before full implementation.
5	Pilot Testing	Controlled rollout	Allow selected staff/users to use the system and report problems before wider adoption.
6	Go-Live	Launch the system	Make the website and application available to the intended users.
7	Support	Monitor adoption	Track issues, collect feedback and provide assistance during the initial implementation period
8	Full Handover	Transfer responsibility	Provide the final system, documentation, credentials and maintenance information to the school.

Training Plan

Training will be provided before the system is officially launched. The training will be role-specific so that users are only taught the functionality relevant to their responsibilities.

Staff Training

Staff members will receive training on:

- Securely logging into the system.
- Managing their profiles.
- Creating and publishing announcements.
- Managing events and schedules.
- Uploading relevant documents.
- Accessing information relevant to their roles.
- Following appropriate security and privacy procedures.

Administrator Training

Administrators will receive additional training on:

- Managing users and roles.
- Managing staff and classes.
- Managing applications.
- Managing announcements and documents.
- Maintaining system information.
- Basic troubleshooting.
- Security and access-control procedures.

Parent Training

Parents will receive a short user demonstration covering:

- Downloading/accessing the application.
- Creating or accessing their account.
- Logging in securely.
- Viewing announcements and notifications.
- Viewing their children's information and schedules.
- Accessing school documents and events.
- Updating appropriate profile information.
- Getting help when experiencing problems.

Implementation Timeline

The change will be introduced in phases to reduce disruption.

Phase	Timeline	Main Activities
Preparation	Week 1	Confirm requirements, identify existing processes and prepare users.
Deployment Preparation	Week 2	Configure production environment and prepare system data.
Staff Training	Week 3	Train administrators and teachers and provide documentation.
Pilot Rollout	Week 4	Selected users test the system and provide feedback.
Full Launch	Week 5	Website and mobile application are made available to the intended users.
Monitoring & Support	Weeks 6–8	Monitor performance, resolve issues and collect user feedback
Final Handover	Week 8	Complete documentation, transfer responsibility and formally close the implementation.

Support and Resistance Management

Some users may initially be hesitant to adopt the new system because they are comfortable with existing processes or may have limited experience with digital systems. To address this, the team will provide demonstrations, training materials and ongoing support rather than requiring users to immediately change all their existing processes.

During the initial rollout, users will be encouraged to report problems through the designated support channel. Technical issues, usability problems and feature requests will be recorded and prioritised. Critical issues affecting access or security will receive higher priority than minor usability improvements.

User feedback will also be collected through discussions, questionnaires or direct feedback from staff and parents. This feedback will be used to identify areas where the system can be improved in future versions.

Handover from Development to Production

Once development and testing have been completed, the system will move from the development environment into the production environment. Before the final deployment, the team will confirm that the major functionality has been tested and that the production environment has been configured correctly. The formal handover to the school will include:

- The completed website and mobile application.
- Production deployment information.
- Database and data-storage documentation.
- User accounts and access information.
- User manuals and training materials.
- Administrator instructions.
- Maintenance and troubleshooting documentation.
- Security and access-control information.
- Backup and recovery procedures.
- Source-code/version-control information where appropriate.
- A record of known issues and future recommendations.

The school will formally confirm that the system has been received and that the required functionality has been demonstrated.

Post-Implementation Review

After implementation, the team will conduct a post-implementation review to determine whether the system has achieved its intended objectives. The review will consider system usage, user feedback, technical issues, performance, and whether the system has improved communication and administrative processes.

Any identified issues will be documented and prioritised. Minor improvements can be included in future updates, while significant changes will be assessed through the project's change management process to determine their impact on time, cost, functionality, and resources.

The change management strategy ensures that the Koos Matli Primary School system is not simply developed and deployed but actively adopted by the organisation and its users. Through stakeholder involvement, phased implementation, role-specific training, user support, controlled migration, feedback collection, and a formal handover process, the transition from existing processes to the new digital platform can take place with minimal disruption. This approach will help the school realise the benefits of the system while ensuring that staff and parents are confident and supported throughout the change.

APPENDENCES

Declaration of authenticity

I, Nkumbu Kalyepe ID Number, 0504281468083 hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

I confirm that we have not committed plagiarism in the accomplishment of this work, nor have I falsified and/or invented experimental data.

I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: ST10446547
STUDENT NAME: Nkumbu
STUDENT SIGNATURE: _____ DATE: 17/08/2026

Declaration of authenticity

I, Moagi Mpinane Theledi ID Number, 0304050082086 hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

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STUDENT NUMBER: ST10026238
STUDENT NAME: Moagi
STUDENT SIGNATURE: Theledi DATE: 17/08/2026

Declaration of authenticity

I, Tlhomamo Tshiamo Tshenkeng **ID Number,** 0304250418080 **hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.**

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I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: ST10349589
STUDENT NAME: Tlhomamo Tshiamo Tshenkeng
STUDENT SIGNATURE: TTshenkeng **DATE:** 17-08-2026

Declaration of authenticity

I, Oratile Matlhodi Lekana ID Number, 0501120372088 hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

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I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: ST10452622
STUDENT NAME: Oratile Matlhodi Lekana
STUDENT SIGNATURE: OMLEKANA **DATE:** 17/08/2026

PROTOTYPE

- **Website Prototype (Canva View Link):** https://canva.link/ovcuch1266h1dnc
- **Mobile Application Prototype (Canva View Link):** https://canva.link/6ino2yc7edalb2d

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